CHAPTER 6

Alternatives

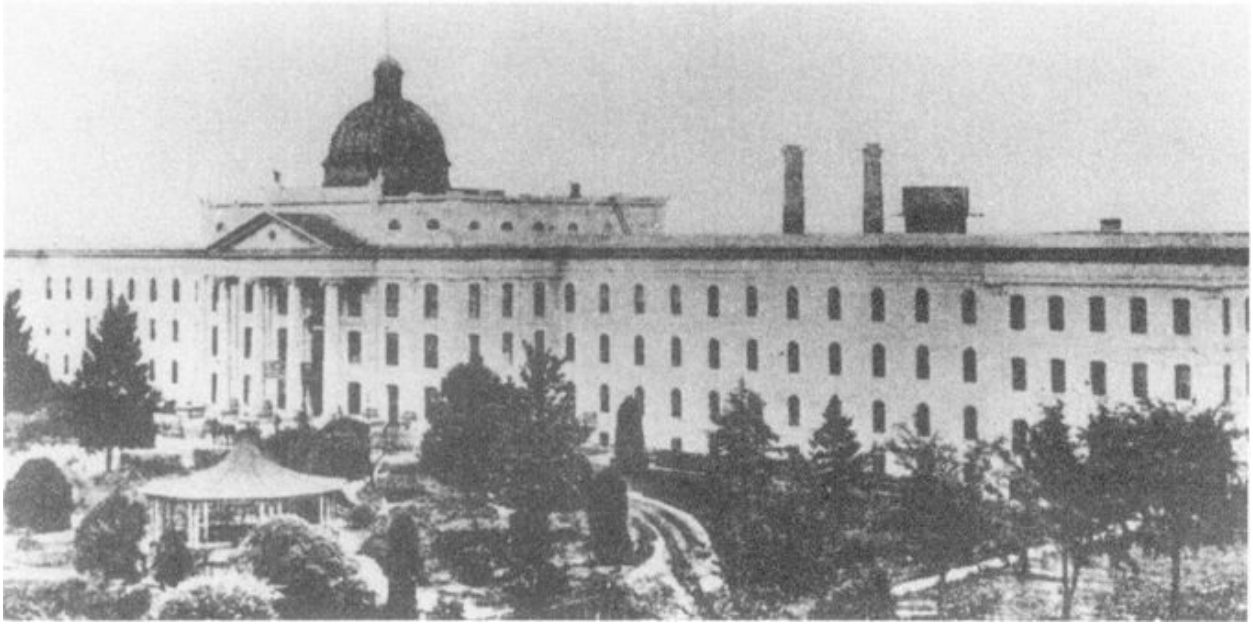
In the first half of the twentieth century, psychiatry was caught in a dilemma. On the one hand, psychiatrists could warehouse their patients in vast bins in the hopes that they might recover spontaneously. On the other, they had psychoanalysis, a therapy suitable for the needs of wealthy people desiring self-insight, but not for real psychiatric illness. Caught between these unappealing choices, psychiatrists sought alternatives. Some of these alternatives proved to be dead ends and were discarded ; others became the basis of a new vision of psychotherapy; still others laid the groundwork for the revolution in drug therapy that would take place after World War II.

At the outset, however, all of these alternatives had an aura of desperateness about them, seemingly radical and possibly quite dangerous innovations. This desperateness must be understood in the context of the time. The asylums were filling, and psychiatry stood helpless in the face of disorders of the brain and mind. In these years, the profession reached the nadir of its descent from the therapeutic promises that had beckoned so brightly a century before. In the 1920s and 1930s, the center of gravity of psychiatry lay in the mental hospitals. In these snake pits, a bleakness prevailed that would have turned away any but the most resolute young medical graduate.

Bodies continued to pile in. Between 1903 and 1933, the number of patients confined in psychiatric institutions in the United States more than doubled, from 143,000 to 366,000. The vast majority were in institutions with more than a thousand beds.[1] There were psychiatric hospitals such as Georgia's notorious Milledgeville with more than eight thousand patients.

Georgia State Sanatorium at Milledgeville, at 10,000 beds by 1950 the largest mental hospital in the U.S., Home of the "Georgia Power Cocktail," a punitive form of electroconvulsive therapy (courtesy of Central State

Hospital at Milledgeville).



Yet as mental hospitals grew in size, they fell in therapeutic power. In England, patients' rate of recovery dropped from 40 percent in the 1870s to 31 percent in the 1920s. At this point, even insiders began asking, "Has scientific psychiatry failed?"[2] Lothar Kalinowsky, a European refugee who in 1940 had joined the staff of the New York Psychiatric Institute, later reminisced, "Today's psychiatrists [1980] do not realize that those of us working in psychiatric hospitals before the 1930s could do little more for our patients than make them comfortable, maintain contact with their families, and in case of a spontaneous remission return them to the community."[3]

These mental hospitals, as asylums became called, seemed the very mirror of desolation. One English psychiatrist likened them to prisons, closed to outsiders and escape-proof. The doctors of course were uninformed in white coats, the nurses in regulation caps. The patients too were uniformed, in "coarse, ill-fitting suits" for the men and drab sacklike dresses for the women. Decorative schemes in the wards and corridors were "confined to variations on two colours—dark chocolate and oleaginous green." Walking through the wards, one would see the schizophrenics "who spent their entire day in assumed statuesque postures ... or in rocking rhythmically and

tirelessly backwards and forwards." In those days smearing with feces and open masturbation were still common.[4]

It had not been all downhill. These 1930-style asylums were considerably cleaner than half a century previously. Discharge rates for younger patients were actually quite high.[5] The horror of true lifelong incarceration was encountered in institutions for the mentally retarded and not in psychiatric hospitals. Yet from the viewpoint of physicians who had to practice in them, asylum psychiatry counted scarcely as a branch of medicine at all. One could cure nothing. There was little scientific understanding of mental illness. And one lived in rustication far from the medical centers with their state-of-the-art labs and great libraries. Younger, often idealistic psychiatrists bridled at this sterile incarceration and sought alternatives.

Fever Cure and Neurosyphilis

Most of the twentieth century has seen restless experimentation within psychiatry to find a cure for the chronic psychoses, mainly schizophrenia and manic-depressive illness. This long search was touched off by the surprising discovery in 1917 of a cure for neurosyphilis, the great middle-class bugbear of the nineteenth century. The story is indissolubly connected with the name of Viennese psychiatry professor Julius Wagner-Jauregg.

Wagner-Jauregg did not feel an inner calling to psychiatry, to say the least. Born in 1857 in the village of Wels—it was always said that he looked like an Upper Austrian woodcutter—he had graduated from Vienna in medicine in 1880. Like most Austrians of his day, he was a dyed-in-the-wool anti-Semite. When a Jewish colleague who had graduated three years previously received a residency with the famous internist Heinrich Bamberger that Wagner-Jauregg coveted, Wagner-Jauregg cursed the entire "Polish Club" and decided to try some other field of medicine.[6] He chanced upon psychiatry simply because the material circumstances of life in the Vienna Asylum were fairly comfortable.

Yet he soon found himself intellectually engaged with the organic side of the discipline. In 1883, during his residency at the asylum, Wagner-Jauregg noted that a female patient who had contracted erysipelas, a streptococcal infection, experienced a remission of her psychosis. This piqued his interest

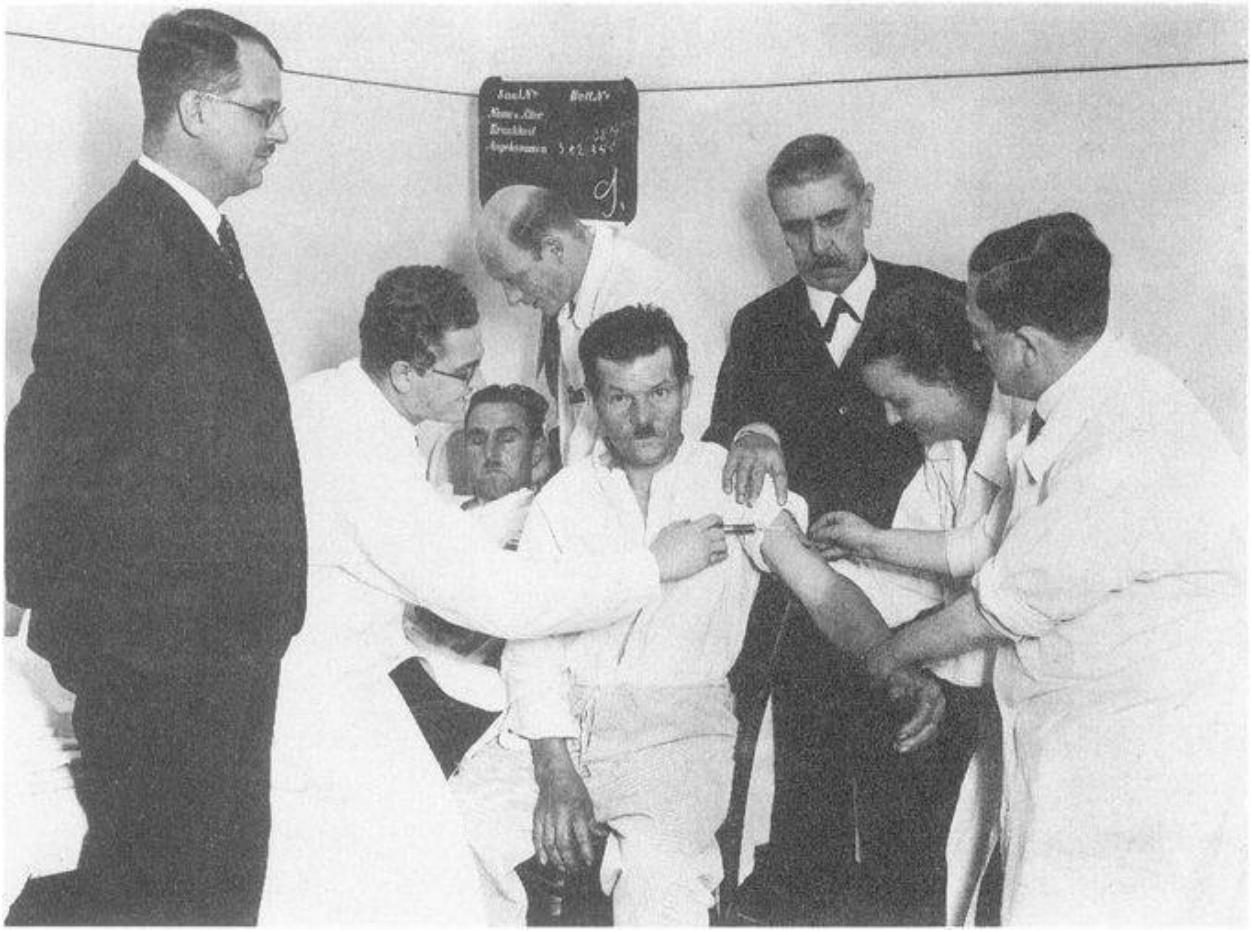
in the relationship between fever and madness, which long had been a subject of medical inquiry. In 1887, Wagner-Jauregg wrote an article speculating that it might be possible to treat psychosis through the use of fever. He mentioned neurosyphilis as being potentially treatable. Later in the article, without particular reference to neurosyphilis, he suggested that inoculating psychotic patients with blood from malarial patients might be attempted.[7] Fate, however, took a hand, for in 1890 the German microbiologist Robert Koch developed a vaccine, tuberculin, that was supposedly effective against tuberculosis. Wagner-Jauregg injected tuberculin into several patients whose psychotic symptoms were caused by neurosyphilis, with the aim of giving them a tuberculous fever. (It was thought that fever itself arrested the progress of neurosyphilis, on the grounds that the syphilis spirochetes are heat-sensitive.) By 1909, he was regularly obtaining long-term remissions of the symptoms of neurosyphilis through the use of tuberculin.[8] Yet he discontinued his experiments with tuberculin because it was considered to be toxic.

Wagner-Jauregg then returned to the possibility of giving paretics a fever with malaria, which, unlike other possible infections, had the advantage of being controllable with quinine. In June 1917, he learned that one of his patients, a soldier sent back from the Macedonian front with shell-shock, seemed to have malaria. An assistant physician asked Wagner if the patient should be given quinine? No, said Wagner, who decided upon the spot to inject some of the soldier's blood into his neurosyphilitics.

In May 1917, a 37-year-old actor with the initials T. M. had been readmitted to the clinic with the now advanced symptoms of neurosyphilis, including weakness of memory, fits, and pupils that were unequal in size and unresponsive to light, a clinical picture that customarily amounted to a death sentence. There being nothing to lose, on June 14, 1917, Wagner-Jauregg inoculated T. M. with malaria. Three weeks later, the patient had his first febrile attack, and after nine such attacks was given quinine. Astonishingly, after the sixth malaria attack, the syphilitic fits came to an end. "In the course of the following months, there was gradual improvement to the point of abolition of all of the patient's symptoms. From August until November, the patient, who at admission to the clinic had been incapable of working, was able to give weekly lectures and musical declamations to the patients of the head-wound clinic ... drawing upon a great repertoire that he recited superbly by memory." T. M. was discharged apparently well on December 5, 1917. A

year later Wagner-Jauregg gave the first report of this work, describing the effects of the malaria-cure upon a total of nine patients.[9] This was an epochal moment not just in the history of psychiatry but the entire history of medicine. Wagner-Jauregg's fever "cure" (it did not cure but it did restore an almost normal life to patients who otherwise would have died demented) broke the therapeutic nihilism that had dominated psychiatry in previous generations. If one could halt the neurosyphilitic psychoses, perhaps psychotic illness from other causes was treatable as well. Wagner-Jauregg received the Nobel Prize for this work in 1927.

Julius von Wagner-Jauregg (on right), Viennese psychiatry professor, looks on as blood that has been taken from the arm of a patient with malaria is injected into the arm of a patient with neurosyphilis. This "malarial fever cure," discovered in 1917, was the first successful physical therapy in psychiatry. Photograph from 1934 (courtesy Institut für Geschichte der Medizin, Vienna).



Wagner-Jauregg's fever cure was tried on every condition imaginable within psychiatry, as for example on schizophrenia.[10] Alas, the fever cure did not turn out to be a panacea.[11] It also had powerful disadvantages. Although it was fairly effective in late-stage neurosyphilis, obtaining marked improvement in up to half of the cases, it was cumbersome, dangerous, and relied upon the availability of people of the same blood type as the paretic patient's. They also had to be infected with the right kind of malaria: tertian malaria producing chills every third day or so. Fever therapy was also very expensive.[12]

Yet Wagner-Jauregg's fever cure was a beginning, touching off other efforts to discover cures for psychosis. For decades to come, the search for physical therapies for the "functional" psychoses, meaning those in which no obvious lesion was present, was the main narrative strand in the history of major psychiatric illness.

Wagner-Jauregg and the Viennese organicists belonged to the grand

Central European narrative of psychiatry's history, on par with Meynert, Griesinger, Freud, and other memorable names. But with the political turbulence of Nazism, efforts to find a cure for mental illness would shift to the New World. Just as the epicenter of psychoanalysis had gravitated to Central Park West, that of organic psychiatry too migrated westward, though not necessarily to New York.

This shift first occurred in efforts to improve on Wagner-Jauregg's fever cure. In 1910, the Berlin physician Paul Ehrlich had announced that the compound "Salvarsan" blocked the development of primary and secondary syphilis. It was an arsenical, a combination of arsenic and an organic substance, offering a way of stopping the spread of syphilis before it reached the central nervous system. Generations of American physicians remembered Ehrlich's discovery as arsphenamine, or "606," standing for the 606th compound that Ehrlich had tried. Yet the general jubilation that prevailed within the medicine about Salvarsan was moderated within psychiatry. To be effective in neurosyphilis, Salvarsan needed to be implemented early. The spirochete that causes neurosyphilis has, however, a long latency period in the central nervous system, and by the time that symptoms became clinically evident, it was often too late.

It was in peripheral American centers, precisely not in psychoanalytically dominated New York, that the definitive end to the neurosyphilis story was achieved with penicillin. The story began in 1929 at Oxford with Alexander Fleming's discovery that mold cultures of penicillin inhibited the growth of bacteria. At the beginning of World War II, Oxford researchers then recognized that penicillin was very useful against bacterial infection at the bedside. The problem was that so little of it was available, and what little could be produced was reserved for the needs of the military. Breaking this bottleneck was a "peripheral" American achievement, first attempted at Peoria, Illinois, and ultimately carried on at twenty-one different drug companies across the country (often using big fermentation vats first installed for making vitamin C to add to orange soda pop).[13]

Linking penicillin to neurosyphilis was very much an American accomplishment. In 1943, John Mahoney, a commissioned medical officer in the U.S. Public Health Service and head of the Venereal Disease Research Center at the service's marine hospital on Staten Island, obtained enough penicillin to be able to try it on primary syphilis, where it proved highly effective.[14] The Office of Scientific Research and Development of the U.S.

government then organized a Penicillin Panel to investigate just what diseases penicillin was good for and how it should be administered. Clinics at eight different hospitals and universities were assigned to study penicillin in neurosyphilis. By August 1944, it was clear that penicillin was an overwhelming success.[15]

Penicillin was tried, for example, on a 34-year-old female paretic at Johns Hopkins Hospital. She had been experiencing auditory hallucinations, disorientation, tremor of her tongue, hands and mouth. By the sixteenth day of her penicillin treatment, she had become completely oriented and was now free of tremor; her speech and writing had returned to normal; the hallucinations had vanished, and she was "satisfactorily performing housework including marketing ... and driving car."[16] The handwriting of another patient, Helen M, had been reduced to a mere squiggle, so completely had she lost the use of her muscles. Penicillin treatment was started March 23, 1944. Helen M's handwriting reemerged like a bather from the depths steadily reacquiring its legibility. By May 13, it had become a proper signature.[17]

These were stunning stories. Neurosyphilis had once filled the asylums. Here was a definitive demonstration that one cause of insanity, at least, was curable.

Early Drugs

As psychotic illnesses go, neurosyphilis was an exception because its cause was clearly infectious. Penicillin would cure no other major psychiatric disorder because few others (aside from febrile delirium) are caused by a virus or bacterium. How about the so-called "functional psychoses" such as schizophrenia, where the cause was unknown? What did the new drug therapies before World War II have to offer these patients? Here the record is more unsettling.

The use of drugs in psychotic illness is almost as old as time. Administering laxatives to the mentally ill on the assumption that toxins bottled up in the colon were making them insane, reached back to the Middle Ages and before. Indeed laxative "cures" of mental illness would remain a constant theme in psychiatry throughout the nineteenth and into the twentieth

century. “Diarrhoea very often proves a natural cure of insanity,” opined Bedlam’s John Haslam in 1809. Haslam called laxatives (“cathartics”) “an indispensable remedy in cases of insanity.”[18] As late as 1921, English psychiatrists were still expounding upon the virtues of croton oil (an oleaginous substance that irritated the bowels and caused diarrhea) “to abort or cut short a mental crisis.”[19] Opium has a history in medicine as old as time and for centuries served the wealthy as a sedative. A young French demimondaine, unable to sleep for anxiety about her emotional life, wrote to her lover in 1773, “Suffering has softened my soul and I yield to it. At five in the morning I took two grains of opium. I obtained from it some calm that was better even than sleep.”[20]

Yet if some themes in drug therapy have remained constant over the ages, other have waxed and waned. The early-modern period had its own therapeutic fancies, such as administering hellebore, or veratrum viride, a plant-derived drug that depresses the heart rate and causes vomiting. Hellebore had gone out by the beginning of the nineteenth century. Instead, two new trends in the pharmacotherapy of insanity established themselves: the widespread use of other kinds of alkaloids (plant-based drugs containing nitrogen); and the rise of compounds used mainly as sedatives and sleeping medications, synthesized in laboratories of the mainly German organic chemical industry.

The first of the new alkaloids to enter the asylum was morphine, isolated from opium in 1806. Like opium, morphine itself had been taken orally for many years. Samuel Kirkbride prescribed it in water for his patients. [21] A new phase began in 1855, when Alexander Wood, an Edinburgh physician, described the use of the hypodermic needle for injecting morphine directly into the circulation. He wrote thus of Miss X, an “old lady” with a long history of nervous symptoms, shoulder pain, and chronic insomnia (who had a history of fainting after taking opium orally): “On November 28, I visited her at 10 P.M. to give the opiate the benefit of the night. Having ascertained that the most tender spot was [in her shoulder], I inserted the syringe ... and injected twenty drops of a solution of murate of morphia.” Within ten minutes she began “to complain of giddiness and confusion of ideas.” Half an hour later, her pain was gone. When he visited her again at 11 A.M. the following morning, “I was a little annoyed to find that she had never wakened; the breathing also was somewhat deep, and she was roused with difficulty.” Her shoulder pain was gone, never again to return.[22] Here was

clearly something new in sedation: an injectable drug that not only calmed patients (a sedative) but put them to sleep for long stretches (a hypnotic). Within asylum psychiatry in the second half of the nineteenth century, subcutaneous injections of morphine became an important means of subduing agitated patients, the practice finally fading when it was realized how addictive morphine was.[23]

The nineteenth century's second half was the "alkaloid period" of asylum treatment. Among numerous alkaloid sedatives, by far the most popular was hyoscyamus, a drug derived from the henbane species of the Solanaceae plant family, a family famous as a source of hallucinogens. Toward the end of the eighteenth century, various authorities had noted that henbane extracts calmed "maniacal delirium." In 1833, chemists isolated the alkaloid hyoscyamine from henbane and the Merck company in Darmstadt began marketing it for various nonpsychiatric indications. Finally in 1868, the Viennese pharmacologist Karl Schroff established that hyoscyamus acted as a sedative and hypnotic. Robert Lawson of the West Riding Asylum in Yorkshire began giving it to his patients around 1875.[24] By the 1880s, hyoscyamine cocktails had come into wide use in institutions. This represented the true beginning of asylum psychopharmacology.

Yet the hyoscyamine story continued to evolve. In 1880, hyoscine, another drug that would later be extensively used in psychiatry (called in the United States scopolamine), was isolated from hyoscyamine.[25] It became an essential ingredient of the sedative "cocktails" used on manic patients, who otherwise would literally agitate themselves to death. As late as the 1930s, psychiatrists at places like the Bethlem Royal Hospital used hyoscine in dealing with "head-banging, rubbing, and pulling out of hair, relentless picking away at the skin causing sepsis.... In extreme form these symptoms, as well as excitement and aggression might be relieved for the time being by Hyoscine Co A, a potent compound of hyoscine, morphine, and atropine [a close relative of hyoscyamine]."[26] Well into the twentieth century, knowledge of these "cocktails" represented the basic lore of asylum psychopharmacology.

The story of hyoscyamine and its relatives shows the typical development of most drugs in psychiatry: They would be discovered and isolated by chemists and pharmacologists interested in other things entirely than disorders of the mind. An inventive psychiatrist would discover that the drug was good for X or Y; then a drug company would purify it and put it on the

market so that it could be given to patients. This was drug discovery largely by serendipity, in contrast to the designed drug discovery of the second biological psychiatry.

During the long "alkaloid period," psychiatry never pretended to anything other than the momentary relief of symptoms with drugs. It is when we come to drugs synthesized in the laboratories of pharmaceutical houses that we encounter efforts, misguided or not, actually to cure patients who have psychotic illness. It is a story that begins largely in Central Europe—because of the enormous organic chemical industry in Germany—and ends in the United States, because of the emigration of Jewish scientists to the New World.

At the heart of this story is the torrent of sedatives that flowed from the workbenches of companies such as Bayer into the asylum. The first of these manufactured sedatives, chloral hydrate, was synthesized in 1832 by the Giessen chemistry professor Justus von Liebig. Liebig represented the first direct tie-in of psychiatry to industry. The founder of organic chemistry, he taught many of the chemists who ended up working for Bayer after it founded a pharmaceuticals division in 1888.[27]

Chloral became the first rehearsal of the "Prozac" scenario (see pp. 320-324) seen with drugs that acquire a great public following for the relief of common psychiatric symptoms. In 1869, Otto Liebreich, professor of pharmacology in Berlin, determined that chloral functioned as a hypnotic to relieve insomnia in anxious and depressed patients who were not insane. Plays and novels of the epoch often eased references to chloral into the plot, the heroine being drugged before being robbed of her virtue. Fans of detective stories recognized chloral as "knockout drops" and "Mickey Finns." Chloral trumped morphine and the Solanaceae alkaloids because it was reliable in strength from dose to dose and didn't need to be injected (although patients disliked its awful taste and characteristic afterodor on the breath). It remained for decades the workhorse of asylum pharmacology and enjoyed wide popularity—and abuse—as a drug that middle-class patients could take at home to avoid the asylum.

Women in particular often became chloral addicts for this reason, treated at home for psychotic symptoms because the family was too embarrassed to have them committed. Theodor Meynert referred a woman of 42 who had become addicted to chloral to a private medical clinic in Vienna for treatment. He thought that addiction was her main problem. But it was not,

for when the clinic doctors took her off the chloral, a psychosis came to the surface. She had to be taken to a private asylum. [28] Virginia Woolf, with a long history of major depression and of treatment in private nervous clinics behind her, was taking chloral at home in the 1920s as she became involved in a sexual relationship with Vita Sackville-West. “Goodnight now,” she wrote Vita in 1928. “I am so sleepy with chloral simmering in my spine that I can’t write, nor yet stop writing—I feel like a moth, with heavy scarlet eyes and a soft cape of down—a moth about to settle in a sweet bush—Would it were—ah but that’s improper.”[29] Vienna and London: two typical trajectories in psychiatric home-care among the middle classes.

Other alkaloids would be almost comical, were the conditions they sought to ameliorate not so desperate. In mania, for example, patients would rage and agitate until they expired. Apomorphine, an artificial alkaloid of morphine that served as a powerful emetic (making patients vomit), came into the asylum after its discovery late in the nineteenth century. It acted against mania. At the Douglas Hospital in Montreal physicians mixed apomorphine with hyoscine, which calmed the patients by making them vomit.[30] At the asylum in Independence, Iowa, “in days gone by, patients who were bothered by mania and could not slow down were administered this drug. They were said to literally turn green and vomit for up to an hour. This would have a sapping effect and they would finally be able to get six hours or so of much needed rest.”[31]

Chloral and apomorphine represented cures only in the sense of suppressing symptoms for a period. The first attempt to cure with drugs occurred in a rather bizarre setting: Shanghai at the end of the nineteenth century. It occurred quite by accident. This story involves the bromides, and requires a bit of background, for the element bromine, which occurs naturally in seawater and salt springs, was first isolated in 1826 from the ashes of seaweed by an apothecary in Montpellier. Believing it to be a substitute for iodine, French physicians began using it immediately on a variety of ailments, for all of which it was unsuitable. Bromine in its natural form is corrosive and had to be administered as a salt, combined with elements like potassium. The French noted that the salts of bromine often produced sedation, an *invresse* bromurique such as the English novelist Evelyn Waugh described in *The Ordeal of Gilbert Pinfold*, a fictionalized autobiographical account of his own experience with bromism.[32]

Bromine entered mainstream medicine in 1857, when Charles Locock, a

London internist with a tony Mayfair practice, was commenting on a paper about epilepsy at a medical meeting. Locock happened to mention that he had given 10 grains of potassium bromide three times a day over a period of two weeks to a patient with "hysterical epilepsy," suppressing the epilepsy. He had also given it "in cases of hysteria in young women, unaccompanied by epilepsy," finding it "of the greatest service."[33] He implied that it sedated them. Away bromium went on its career as a sedative.

The bromides spread rapidly within public asylums because they were cheaper than chloral. By 1891, the Paris asylums were using over a thousand kilos of potassium bromide a year.[34] Older asylum psychiatrists still remember giving "triple bromides ... in liquid form to thousands of patients q. h. s. [at bedtime] and frequently p. r. n. [as needed] during the night."[35] The point is that when, around 1879, young Dr. Neil Macleod went out to Shanghai to practice medicine, he would have been thoroughly familiar with potassium bromide and perhaps already confident in its application.

Prolonged Sleep

Macleod, a recent Edinburgh medical graduate, quite unwittingly became the innovator of the first drug therapy in psychiatry that actually promised relief of psychotic illness.[36] Early in 1897, a married Englishwoman of 48, resident in Shanghai but staying in a hotel in Japan, "received by wire unexpected news of a family affliction, giving rise to a great nervous shock." The sounds of the other hotel guests soon began to drive her "frantic." She developed an attack of acute mania, and Macleod was summoned from Shanghai to bring her home. Not having skilled nurses available—but possessing considerable experience with weaning morphine and cocaine users from their addictions by putting them into prolonged bromide sleeps[37]—Macleod decided to try the same method on her that he had used on the addicts, even though her problem was psychosis. He gave her a large dose of bromide and then after she went into a deep sleep had her transported out in a hammock. After a 500-mile steamer trip and an odyssey lasting several days, she finally arrived back in Shanghai, awakening well without "a trace of mental disturbance."[38]

She remained well for two years, then relapsed in June 1899 after “another shock” that left her unable to ride to the hounds. She became irritable, then passed into “a state of incoherence, exaltation, and incessant speech activity with delusions of fear in the night and the desire to escape from the house, neither it nor its inmates being recognized.” Macleod again diagnosed acute mania. In the absence of “a padded room” or “skilled attendants” within thousands of miles of Shanghai, he decided to treat her again with what he was now calling “the bromide sleep.”

On the first day, Macleod gave her sodium bromide in two-dram doses (about a quarter of an ounce) starting at 8:30 in the morning and continuing for three more times that day. She wandered about the house for most of the day “talking nonsense nearly incessantly” and fell asleep about nine that evening. The second day, he gave her another ounce. By the end of this day, she was sleeping quite a bit and was “full of delusions expressed quite distinctly.” At this point, he stopped the administration of the bromide. She had swallowed two full ounces. The third day, she could speak only with difficulty and could not walk without being supported. The fourth day, she could not be roused. Nor could she be roused for the next three days. On the ninth day, she began to surface, withdrawing her foot, for example, when tickled. Yet only on the twelfth day did she even begin to mutter, and not until another week had passed was Macleod able to assess her mental status: “Twenty-third day: A good night; the faintest trace of mental disturbance. From this day, no departure from the normal mental state could be detected; walking in the garden and downstairs, meals as usual.”

Macleod treated eight other patients as well with his “bromide sleep.” One died of pneumonia, the likely result of having aspirated vomitus while in a stupor. The others had either been cured of addiction or, like the preceding patient, of “acute mania.” As Macleod described his newly discovered sleep, “the patient slept day and night for five to nine days, unable to “walk, stand, sit, speak, or carry on any of the higher cerebral functions.” Every six hours, Macleod would put the patients on a commode, and every few hours give a tumblerful of milk. Upon awakening, the patients seemed actually to have recovered from their illnesses. It is unknown how many relapsed.[39]

The profound duration of torpor differentiated Macleod’s deep-sleep therapy from previous psychiatric efforts to narcotize patients for brief periods.[40] For the first time in the history of psychiatry, a drug therapy had been described that seemed to alleviate major psychiatric illness with a

physical procedure. Whether Macleod's bromide sleep really did cure patients is beside the point: A hint had been inserted into the profession's collective thinking that some kind of cure with drugs might be possible.

The bromide sleep cure was taken up briefly by other physicians, then abandoned.[41] Bromine itself was perhaps too toxic, or the whole notion too recklessly "heroic" to contemplate. The next step in the development of sleep cures would not occur under the aegis of bromine but of the barbiturates.

In 1903, the German chemist Emil Fischer and his collaborator Joseph von Mering modified a class of drugs originally synthesized in 1864 in a way that made them effective as sedatives and hypnotics.[42] The original inventor of the compounds had called them barbiturates, apparently after his girlfriend Barbara. Fischer and Mering realized that their new drug, "diethyl barbituric acid," or barbital, was a sedative. It improved vastly upon the congeries of previous sedatives by not tasting unpleasant, by having few side effects, and by acting at therapeutic levels far beneath the toxic dose (unlike potassium bromide, which tasted awful and had a therapeutic level close to the toxic dose). The Bayer company marketed Fischer's discovery as "Veronal," the Schering company as "Medinal." Both brand names would become household words.[43] The barbiturates were pricey. In Germany, one gram of Veronal cost 40 pfennigs, one gram of Trional, a competing nonbarbiturate sedative, 15 pfennigs.[44] Nonetheless Veronal and similar barbiturates caught on immediately. They calmed maniacal patients, restored the sleep of the melancholic, and were an effective hypnotic for insomniacs in general (and still are: the Lilly Company's short-acting Seconal, or secobarbital, is commonly used in North America today).

Veronal was first tried on patients in 1904 by Hermann von Husen, a young asylum psychiatrist who himself had problems sleeping. Among his other findings, he reported that one evening he had taken half a gram of Veronal, the next evening a gram. "In both cases after 10-15 minutes I felt an ever-growing weariness, which went over to deep sleep after half an hour. After half a gram of Veronal I slept 8 hours, after a gram around 9 hours. The first morning I awoke fresh and rested, the second morning after the higher dose I had trouble getting out of bed...."[45] No other drug then available let people sleep deeply then wake up refreshed. The potential appeal of such a product was immense.

Veronal became the drug of choice in private nervous clinics (while the public asylums stayed with cheaper standbys such as bromium and chloral).

Here is “Jane Hillyer,” a young woman in the grip of manic-depressive illness, during one of her stays in a sanatorium:

> The sun was just beginning to sink behind the trees.... It would be hours before it was time for sleep—for Veronal. I continued my imperious demands. “Maybe they will break their old rules; maybe I can have it earlier if I make a fuss”; they did, a little. Yet long before the nurse stood before me with a glass of clear water and that magic white tablet—it always seemed the one cool, white thing in the world—long before then, I had heard my own voice, cracked and strange mingling with the unearthly cries of other patients....“[46]

Dozens of other barbiturates came onto the market after the success of Veronal. In 1912, Bayer marketed Luminal (phenobarbital), a barbiturate still used for epilepsy. Because of its long-acting nature, “phenobarb” became a favorite drug in asylums well-off enough to afford it. “The prevailing anxiolytic,” one elderly psychiatrist recalled, “was elixir phenobarbital. It was pink in color and many patients, most commonly female, carried their bottle of pink medicine with them and took a teaspoonful when necessary.” [47] Not just in asylum psychiatry but within medicine as a whole, the barbiturates became widely used, the family doctor’s familiar recourse in dealing with “a hysterical girl on her bridal night.”[48] If a single class of drugs came to exemplify psychiatry before the advent of Valium (the class of benzodiazepines) in the 1960s, it was the barbiturates.

In this context of growing familiarity with barbiturates capable of inducing prolonged deep sleep, the first major attempts of the twentieth century to cure psychotic illness were launched. The actual priority for inducing prolonged sleep with barbiturates belongs to Giuseppe Epifanio in 1915, an assistant physician at the Turin university psychiatric clinic.[49] Yet his article about therapeutic success with deep sleep, appearing as it did in the middle of a war and in a language that few foreign psychiatrists read, made little impact. Prolonged sleep therapy, the first of the so-called physical therapies of the twentieth century, is associated with Swiss psychiatrist Jakob Klaesi, a staff physician at Zurich’s university psychiatric clinic, the Burghölzli. In 1920, Klaesi began using an innovative combination of two barbiturates of the Swiss drug firm Hoffmann-La Roche marketed as Somnifen. It was Klaesi’s aspiration to cure no less a disease than schizophrenia with prolonged narcosis *(Dauernarkose).*

Klaesi, then 37 and a private docent at the Burghölzli, was by no means a pill-grubbing biological type. He had always been enthusiastic about psychotherapy and indeed argued that deep sleep was a way of calming patients down to make them accessible to dialogue. But he was not known for good judgment—he became a Nazi-sympathizer in the 1930s—subject as he himself was to manic-depressive personality swings. He was also highly ambitious. All of these qualities made him perhaps overly receptive to the suggestion of the Zurich pharmacology professor Max Cloetta that Epifanio's deep-sleep therapy might be tackled again, this time with the aid of Somnifen.[50] "I decided upon introducing prolonged sleep, that is prolonged narcosis, into psychiatry," Klaesi rather grandly began his report, in an effort to "achieve an improved rapport between doctor and patient." Blocking out stimuli through sleep might help dig the schizophrenic patients out of their negativistic attitudes so that psychotherapy could begin. As Klaesi took over the ward for agitated females at the Burghölzli in April 1920, he encountered a 39-year-old woman who had been an "able businesswoman" until falling ill three years previously. She had become anxious, complained that someone wanted to kill her and her husband, had started hearing voices, and was put in a private nervous clinic. She improved, went home, became ill again, and was admitted this time to the Burghölzli in March 1919. She became so violent that the hospital had to put her in a special padded cell that had been unused for two decades.

Klaesi tried hard to establish a therapeutic rapport with her. On one occasion as she lay naked in her cell, Klaesi asked her, what are you "as an intelligent woman doing in such a hole."

"What do you want?" she asked him.

"To cure you at any cost so that you don't have such a horrible life," he replied. Thereupon she began to cry, then attempted to conceal her tears and got a runny nose. Klaesi handed her his handkerchief. Her face opened up. She looked at him smiling and said to the nurse standing by, "He's OK." Klaesi became intent upon establishing a psychological rapport with her.

At the end of April he decided to try and "dismantle her defenses" with a Somnifen course of deep sleep. He followed this by two further courses, each lasting typically five to six days. In October 1920, she was discharged as vastly improved. The husband later said he had never seen his wife "to be so industrious, circumspect and tender as she now is." The woman, however, just before discharge declared to Klaesi that her marriage was in trouble. She

scorned her husband as a Caspar Milquetoast for permitting her to dominate over him—and expressed a desire for the husband to be like her doctors. Klaesi thoughtfully concluded, “Prolonged sleep and Somnifen had opened the way to a fruitful dialogue with the patient and to new insight.”[51]

Of the 26 patients Klaesi treated with prolonged narcosis, about a quarter to a third improved so greatly that they could be discharged or transferred to wards for less agitated patients. On the face of it, this seemed to be a promising new therapy.[52] The problem was that three of the patients had died as a result of pneumonia or circulatory collapse. The Achilles’ heel of prolonged sleep was its high risk. Several years later, Swiss psychiatrist Max Müller at the Münsingen asylum, determined to break out of “paralyzing therapeutic resignation toward my patients,” did another study comparing two barbiturates, and found that prolonged sleep with Somnifen had a mortality of about 5 percent.[53] Klaesi had dismissed the deaths in his own study as owing to preexisting organic conditions, and never forgave Müller for having called attention to the potential lethality of the new treatment for schizophrenia that Klaesi had supposedly discovered.

In the following years, prolonged sleep with safer barbiturates than Somnifen became the first asylum therapy that offered any hope.[54] Deep sleep became widely adopted for affective, or mood, disorders, often by physicians who shared Klaesi’s hope of somehow making these gravely ill patients accessible to psychotherapy.[55] Eliot Slater of the Maudsley Hospital called sleep therapy “the one treatment we had in the early thirties which was of any avail with acute psychotic illnesses.” Maudsley doctors would have their patients sleep 12 to 16 hours a day, “with drowsy intervals in which the patient could be roused to eat and drink.” Afterward, the patient would remember little of the psychotic episode and might expect early discharge from the hospital.[56]

The notion of addressing the brain with profound narcosis spread through psychiatry. German physicians in the early 1930s used it to withdraw addicted patients from morphine.[57] Washington psychoanalyst Harry Stack Sullivan would intoxicate his patients for “three to ten days” with alcohol in order to open them up to psychotherapy. (“Recourse is had to chemotherapeutic agencies, notably ethyl alcohol, which impair the highly discriminative action of the more lately acquired tendency systems ...,” Sullivan explained.)[58] Kalinowsky called Klaesi’s prolonged sleep therapy

“the first treatment with which at least transitory improvement was obtained in functional psychoses.”[59] Thus prolonged sleep is partially exculpated from its condemnation at the hands psychoanalytically oriented historians of psychiatry. It gave new hope to physicians whose life mission was the care of patients with major psychiatric illnesses. After decades of warehousing patients, sleep therapy offered the prospect of cure.[60]

D. Ewen Cameron, professor of psychiatry, McGill University, 1943-1964, a distinguished North American psychiatrist of Scottish origin, whose work ran off the rails in the notorious “depatterning” experiments with sleep therapy and electroconvulsive therapy that he conducted in the 1950s (courtesy McGill University Archives).

Yet this story has an unpleasant codicil. Just as most of the stories of the physical therapies began in the Old World and ended in the New, so did deep-sleep therapy begin in Zurich and end in Montreal, at the hands of Dr. D. Ewen Cameron. Cameron was born in 1901 in Scotland and graduated MB from Glasgow in 1924. Trained at the leading psychiatric centers of the day—Bleuler's Burghölzli and Meyer's Phipps Clinic—in 1929 he started working at an asylum in Canada's province of Manitoba, went on to several American posts, then in 1943 became the first director of Montreal's Allan Memorial Institute (which is the joint department of psychiatry of McGill

University and the Royal Victoria Hospital, housed in a mansion next to the hospital belonging to the former shipping lord Sir Hugh Allan). By the early 1950s, Cameron had become a distinguished researcher, serving for example in 1952-1953 as president of the American Psychiatric Association.

An enemy of psychoanalysis, Cameron grasped at the new physical therapies. Having been in Zurich in the 1920s, what could be more natural for him than to apply in Montreal Klaesi's barbiturate narcosis. Yet Cameron added a twist, in the belief that forcing patients to listen to propagandistic messages ("brainwashing," he called it) during their prolonged sleep would somehow accelerate their recovery. He first attempted this "psychic driving" in 1953.[61] "The [tape] loop was broadcast from a speaker on the wall of the sleep room in a female and (he hoped) maternal voice," reports one writer. "At first [the young female patient from Bermuda] showed little response to the voice. Then, unexpectedly, she became as hostile as her condition would allow: 'at moments when she roused slightly from her sleep, [she] would crawl or stagger out of bed, and try to destroy the source of the voice.' The hostility peaked in about six days. By the end of ten days, 'she not only became quite undisturbed by the voice, but she slowly began to incorporate the content, saying, Yes, I want to go back to Bermuda; yes, my parents love me.'"[62]

This was Cameron's point of departure. In 1955, he began to put patients into deep sleeps, simultaneously giving them electroshock several times a day (passing electricity through the brain to induce therapeutic convulsions). He called this "depatterning," breaking down previous brain constellations that, he thought, had somehow produced psychosis.[63] Over the next decade, his therapeutic judgment clearly became unbalanced as he attempted this systematic depatterning of patients without their consent and without any pretense of applying the scientific method. In 1964 Cameron fled Montreal for a post at the Albany Medical School. He died, essentially in disgrace, three years later. (At his death the New York Times obituary headlined: "Led Research in Geriatrics at Hospital in Albany."[64]) With Cameron, sleep therapy died as well. By the 1960s, many safer and more effective treatments for psychotic illness had come along. Yet it had marked the beginning of a revolution in psychiatry.

Shock and Coma

Why shocking the brain to the point of eliciting convulsions makes psychotic patients better is unclear. But it does. So does the dangerous procedure of putting them into prolonged comas, as opposed to the stupors that bromide sleep elicited. So little understood is the relationship between brain and mind even now that the mechanism of these treatments, which have helped retrieve many patients from the depths of profound illness, is not understood. Yet empirically, some of these therapies have proven themselves, so much so that electroconvulsive therapy, or ECT, is today the treatment of choice for major depression. Their advent in the 1930s, as clearly superior alternatives to either warehousing or to psychoanalysis, marked a great turning of the page in the history of psychiatry.

Manfred Sakel on left, discoverer in 1933 of insulin coma therapy. In the middle is Vienna psychiatry professor Otto Poetzl, in whose clinic Sakel did his early work; on the right is Hans Hoff, head of Vienna's university psychiatry clinic after World War II. Here Sakel is receiving an award in 1957, the year of his death (courtesy of AP/Wide World Photos).

To this day, however, both coma and shock have remained controversial. The antipsychiatry movement has rejected them as a matter of principle, as have physicians who rely on psychotherapy to the exclusion of most else. The historical record, however, shows that coma and shock gave psychiatrists powerful new therapies in a field dominated for half a century by nihilistic hopelessness, and it is against that sense of despair that they must be set.[65]

This story begins in Berlin with a young medical graduate of the Vienna University named Manfred Sakel. Sakel was born in 1900 in Nadverna, Galicia (then part of Austria), of a pious Jewish family said to be descended from Maimonides. By the time of his graduation in 1925, so scorching had the blast of Austrian anti-Semitism become that he found a job in Berlin as assistant physician at Kurt Mendel's expensive suburban private clinic, the Lichterfelde Sanatorium. The clinic courted the actresses and physicians who typically were at risk of morphine addiction. Yet a cold-turkey withdrawal often entailed symptoms such as vomiting and diarrhea. In the late 1920s, Sakel discovered that such symptoms could be successfully managed by administering small doses of insulin, a hormone just discovered in 1922.[66]

Insulin had already been tried several times in psychiatry during the 1920s, as early as 1923, when the staff of the Psychopathic Hospital in Ann Arbor, Michigan, gained the impression that insulin relieved the depression of diabetic patients as well as their diabetes. In fact it did not.[67] Later in the 1920s, insulin was used on patients who had lost their appetites or refused to eat.[68] But it occurred to no one that a coma from insulin shock might be curative. (The hormone causes the muscles to take up glucose from the blood. If too much glucose is withdrawn, the patient will go into a hypoglycemic coma.)

Sakel was probably not aware of much of this previous writing in any event. Yet he did have to cope with the insulin comas that happened inadvertently to several of his own patients. He noted that after the comas were over the patients' desire for morphine had been abolished. In addition, these previously "restless and agitated" patients had become "tranquil and accessible." Sakel reported this finding in 1933.[69] At this point, he was clearly thinking that putting patients into an insulin coma might in and of itself be a cure for major psychiatric illness.

Evidently as a result of the Nazi takeover, Sakel returned to Vienna in 1933, getting a post at the university psychiatric clinic under Otto Poetzl, Wagner-Jauregg's successor as the professor of psychiatry. Sakel also became chief physician of a private clinic in a Viennese suburb. Sakel persuaded a reluctant Poetzl to try out the dangerous-sounding therapy, and in October 1933, Sakel began testing systematically at the university clinic his theory that insulin "shock" represented a cure for schizophrenia.

Colleagues at the clinic were astonished at the results. Said Karl Dussik: "As I worked from the first day in using this method in our clinic I can confirm its real effect. The personality of the patient can often be changed so entirely during the hypoglycemic reactions that it seems as though the glycopenic [low blood sugar] treatment has created a new being."[70] (Success in psychiatry often gives the impression of creating a new person, witness the huzzahs surrounding Prozac many years later, see pp. 320-324.)

In 1934, Sakel broadcast the first results of his study: Inducing hypoglycemia with insulin and putting patients into insulin comas were producing astonishing alleviations of symptoms.[71] Of his 50 patients experiencing their first episode of schizophrenia, he obtained a full remission in 70 percent and a "social remission" in a further 18 percent.[72] The first page of his clinical notebook in Vienna shows 8 of his 12 patients discharged to

their homes as improved. A further patient died shortly after going home, three others were transferred to the Vienna city asylum at Steinhof.[73] These were recoveries in a patient population previously considered hopeless. One notes that at this stage the insulin "shock" did not include convulsions (or if they did, they were considered undesirable side effects).

Yet beyond the university clinic, Sakel's results were considered a joke, the man himself a charlatan, and Poetzl's patronage of him a mystery.[74] Sakel spent only three years in Vienna, traveling to the United States in 1936 to treat a wealthy private patient and staying on, first as a staff member of a New York State psychiatric hospital, then in private practice in Manhattan. It is incredible that the psychoanalytically dominated American Psychiatric Association at first refused his application for membership, admitting him only after the science writer of the New York Times called attention to the story.[75] Sakel died in 1957, having spent his life zealously defending his reputation and his priority.[76]

Rather than being adopted in Central Europe, insulin-coma therapy was taken up in Switzerland and in the Anglo-Saxon world. The Swiss private and public psychiatric hospitals were at the time probably the most progressive in the world, and it is no surprise that in 1937 the director of the Münsingen public asylum, Max Müller, arranged a world conference on new therapies for schizophrenia. Münsingen became the "world mecca" of insulin treatment. "People trust me, not Sakel," Müller boasted.[77]

As with all these new physical therapies, the diffusion of insulin-coma to the non-German-speaking world occurred within a web of personal to-ing and fro-ing. Herbert James Pullar-Strecker, who brought it to Britain, had been born in Würzburg of a German physician-father and a Scottish mother whose maiden name was Pullar. He emigrated to Britain in the 1920s and had just qualified in Glasgow as David Henderson put him in charge of the future insulin unit at the Royal Edinburgh Hospital for Mental Disorders. Pullar-Strecker initiated coma therapy there early in 1936.[78] Later that year Dr. Isabel Wilson of the mental-hospital Board of Control visited Vienna and Münsingen, returning to England with a positive report of the new therapy. She called the burden of schizophrenia "so heavy that it demands the full and careful trial of any therapeutic measure which claims to relieve or diminish it."[79]

For British psychiatrists, who had been suspicious of deep-sleep therapy,

insulin coma came as heaven-sent. Eliot Slater had come to the Maudsley in 1931. He recalled the patients with chronic depression in the preinsulin days: "The involutional melancholic would be a thin, elderly man or woman, inert, with the head lifted up off the pillow. There were some sort of Parkinsonian-like qualities, mask-like face sunk deep in misery, and speaking in a retarded way. If you could get them to say anything, it would be something about how hopeless things were, how they were wicked, doomed to disease, death, and a terrible afterlife, if there was one." Insulin coma was the first therapy that actually permitted doctors at the Maudsley to help these patients.[80]

A sense of the enthusiasm of British hospital psychiatrists for insulin-coma therapy may be gleaned from the letters home of a young Canadian psychiatrist who was doing a "locum" in the summer of 1937 at the Warwickshire and Coventry Mental Hospital in Hatton near Warwick. Writing in July to his chief in Toronto, he said:

> [This] 1,400 bed hospital of block type [has a] staff of six physicians all under 35 and all keen. [The hospital employed a full-time pharmicist and (thank God, in the writer's view) no psychologist.] Our trust is not in extra staff but in drugs and the barbiturates in particular [for mania and depression].... And now of course insulin shock for the schizophrenic group is giving us a chemical procedure for the other great biogenic group. How the heart of the "druggist" has been gladdened these past four years! Dr. D. N. Parfitt our chief spent six weeks in Vienna and we started in cases 5 weeks ago. Only 3 or 4 places in Great Britain have started and no literature of importance has appeared here.... I expect to spend August and September in the insulin shock clinic.[81]

The Warwickshire asylum was in the vanguard by only a bit. "By 1939 every self-respecting go-ahead hospital had its insulin unit," said one observer. [82] In William Sargant's and Eliot Slater's influential 1944 textbook of physical treatments, based on their experiences at the Maudsley Hospital, insulin-coma therapy was clearly the first choice.[83]

Not Sakel but a young American psychiatrist, who happened to be exposed to insulin therapy in 1934 while in analysis with Freud, introduced insulin coma therapy to the United States. Born in 1906, Joseph Wortis had graduated from the University of Vienna in 1932. He went to New York to intern and train in psychiatry at Bellevue Hospital with the recently arrived Viennese psychiatrist Paul Schilder. In the fall of 1934, in the context of his

interest in human sexuality, Wortis underwent a kind of didactic analysis with Freud. During this analysis, in December 1934, he happened to visit Poetzl's psychiatric clinic at the Vienna General Hospital and saw Sakel administering insulin therapy. The enthusiastic young Wortis turned his back on psychoanalysis, returned to Bellevue and initiated insulin-coma therapy in the United States.[84] In November 1936, he gave a paper on the subject that later appeared in a neurology journal.[85] (The report was delayed 18 months, he said, because he could not persuade any of the psychoanalytically dominated psychiatry journals to publish it.[86]) Wortis translated Sakel's monograph into English.

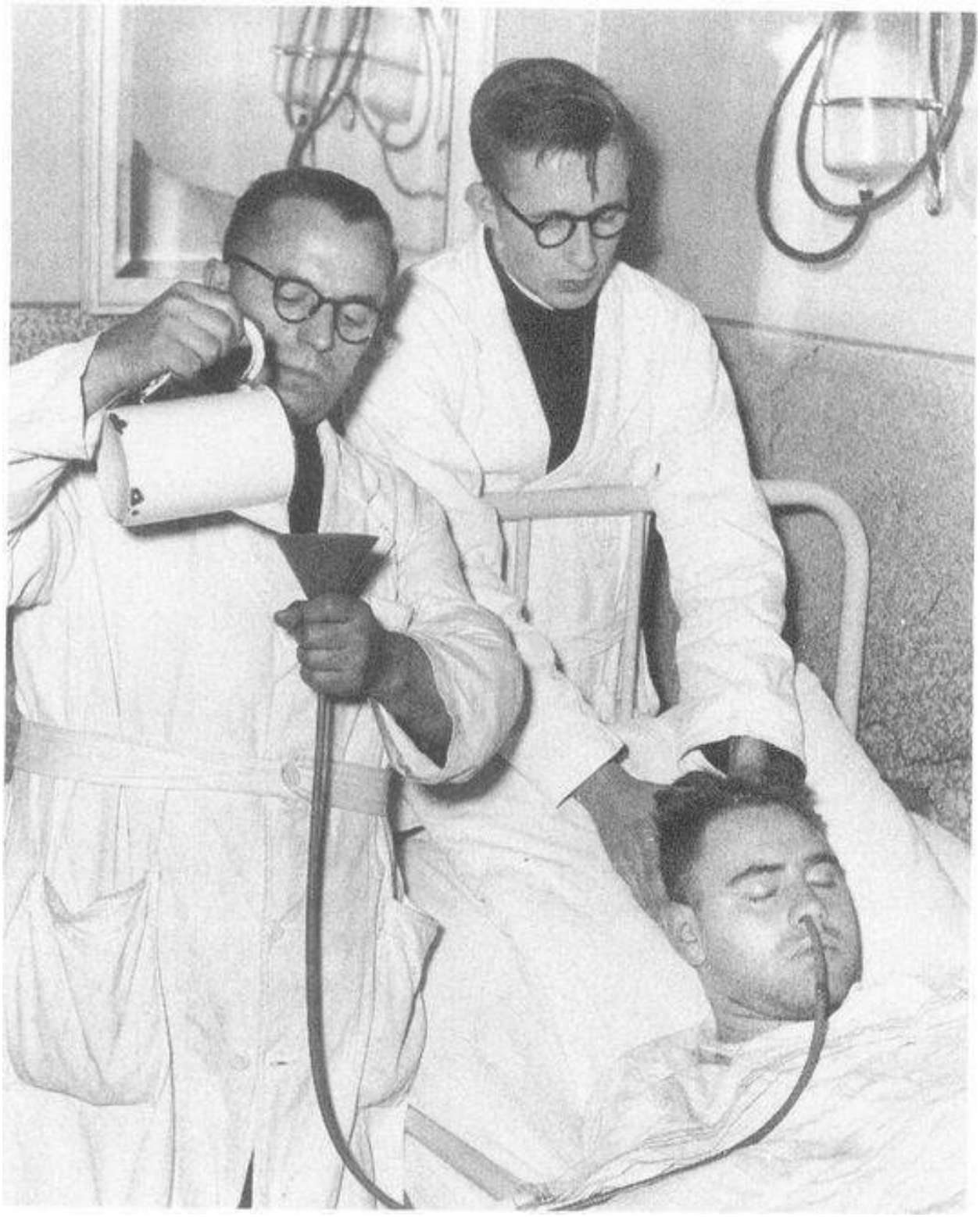
Yet Sakel himself provided his own authoritative imprimatur to the introduction of his therapy in the United States, lecturing on it to a New York medical audience in January 1937.[87] The meeting was an American psychiatric microenvironment: Adolf Meyer was there and gave insulin coma "a verbose blessing," as he gave everything his blessing regardless how absurd. Unsurprisingly, Ewen Cameron made a "forceful contribution advocating insulin coma therapy." Psychoanalyst Smith Ely Jelliffe, in an involuntary caricature of Freudian thought, opined that insulin coma might work by "withdrawing the libido from the outside world and fusing it with the death impulse for the maintenance of the narcissistic ego."[88] Analysts like Jelliffe saw asylum psychiatrists as dim country cousins. Yet in the 1940s and 1950s, these country cousins nonetheless seized on insulin therapy to break out of the therapeutic impasse of custodialism and draw themselves closer to medicine. By the early 1960s, there were special insulin units in more than a hundred American mental hospitals.[89]

How was insulin-coma therapy administered? It was a dangerous procedure with a mortality of almost one in a hundred, and required a team of doctors and nurses in a special unit. There might be "twenty or more patients on the insulin ward all in coma at the same time," said Walter Freeman (the later devotee of lobotomy) of George Washington University. "There was no red flag waving on the brink [of hypoglycemic death]; and most therapists hesitated to approach it too closely."[90]

The therapy would begin in graded little steps. At the Maudsley Hospital said Slater, "Every day [the patient] received a dose of insulin, increasing until he started drifting off to sleep, insulin sopor, and then eventually from sleep into coma, deep unconsciousness." In coma, the patient was watched

constantly and permitted to spend no more than twenty minutes in that condition before being brought round with a sugar solution. (At other hospitals two-hour comas and longer were not unusual.) “When he had had twenty comas or so one usually saw a remarkable improvement in the mental state, at first in the morning after treatment, and then persisting into the rest of the day, and finally persisting after the treatment had been tapered off.” Insulin coma doubled the proportion of recoveries at the Maudsley, said Slater.[91] At Nova Scotia Hospital in Canada, where Charles Roberts ran the insulin unit around 1943, they would steadily increase the insulin doses over a period of 10 days or so until the patients reached the stage of deep coma with occasional convulsions. ”On termination of coma, patients frequently had what was described as a ‘lucid period’ of varying duration during which conversations would be normal and delusions and hallucinations seemed to be absent.“[92]

Brothers of Charity at the Dr. Guislain Asylum in Ghent, Belgium, bringing patient out of insulin coma therapy, c. late 1940s or early 1950s. The man with the rubber tube in his nose has been given a glucose (sugar) solution to bring him out of the coma. A strong concentration of glucose is poured directly into the stomach to prevent the coma from becoming lethal (courtesy of Museum Dr. Guislain, Ghent, which forms part of the present psychiatric centre Dr. Guislain).



Is it any wonder that asylum psychiatrists became keen about insulin coma? It seemed to be a procedure that actually worked, at least for the short

term, without the extreme dangerousness of sleep therapy. In the long term, it was discovered that insulin coma had about the same success rate as barbiturate-sleep therapy.[93] Both represented a substantial improvement on what was available before, which is to say, nothing.

Within months of the adoption of insulin coma, a second so-called shock therapy came along that marked the true beginning of convulsive therapy. The difference is an important one: this second therapy, with Metrazol, produced convulsions without coma. Shocking the brain to the point of eliciting a convulsion does seem to have a beneficial effect upon psychotic illness, particularly on major depression. The era of convulsive therapy began not with insulin (in which convulsions were undesired and incidental) but with a drug similar to camphor called by the trade name Metrazol (Cardiazol in Europe). Its discoverer was Ladislas von Meduna, a 38-year-old Budapest psychiatrist who in 1934 proposed ameliorating the symptoms of schizophrenia by deliberately putting patients into a convulsive state.[94]

Patient in the midst of an unrestrained Metrazol convulsion in an American mental hospital, c. 1941. Discovered in 1934 by Ladislas von Meduna, Metrazol represented the first of the true convulsive therapies.

Source: American Journal of Psychiatry, 97 (1941), p. 1052, copyright 1941 American Psychiatric Association, reprinted by permission.

Meduna had trained as a neuropathologist between 1923 and 1926 at Budapest's Interacademic Institute for Brain Research, then followed his chief Karl Schaffer to the university department of psychiatry in Budapest where Schaffer had just been made professor. Here Meduna encountered clinical psychiatry for the first time. He became intrigued at the changes that his co-workers in the psychiatry department were finding at autopsy in the brains of schizophrenic patients who had died suddenly. As Meduna worked in the outpatient department, he discovered (or believed he discovered) that the brains of patients with epilepsy were quite different from the schizophrenic brains. Meduna pondered this. Then in 1929 other scholars reported that epileptic patients who developed schizophrenia seemed to experience less epilepsy. Meduna wondered if the relationship went the other way as well: Did schizophrenics improve after developing epilepsy? (The answer seemed to be yes.[95]) "I accepted the concept of a biological

antagonism between the two diseases," said Meduna in his autobiography. [96] Could one therefore improve the symptoms of schizophrenia by giving the patients epileptic fits? Meduna selected camphor, a drug with a long history of causing seizures, as the drug of choice. Since the eighteenth century, camphor had been occasionally used in psychosis, though Meduna seemed to be unaware of this.[97] On November 23, 1933, the same day that Sakel gave his verbal report on insulin-coma therapy to the medical society in Vienna, Meduna began animal experiments with camphor. On January 23, 1934, Meduna gave camphor to the first patient.[98]

L. Z. was a 33-year-old man who had been admitted to the Budapest State Hospital in 1930 with the (delusional) belief that "people often wave at me." He heard voices in his ears and his stomach. The entire year 1933 he had spent under his bedcovers. By January 1934 he had stopped eating. On January 23, Meduna gave him the first injection of camphor. Forty-five minutes later, L. Z. had his first epileptic attack, his pupils dilated to the maximum. Over the next two weeks, he would have five more injections. "On the morning of February 10, the patient spontaneously arises from bed, is lively, speaks, and asks for something to eat. He is interested in everything going on about him, asks about his illness and realizes that he has been sick. He asks how long he has been in the hospital, and as we tell him that he has already been there four years he cannot believe it."[99]

That was the published version. In reality, the patient felt so good that he escaped from the institution, went home, "and found out that the cousin living with his wife was not a relation at all but his wife's lover. He beat up the cousin and kicked him out of the house; proceeded to beat up his wife and told her that he ... preferred to live in the state mental hospital where there is peace and honesty."[100]

"From that time on," said Meduna later, "I considered this patient cured." The patient was still well at the time that Meduna left Europe five years later.

By January 1935, as Meduna submitted his first article on convulsive therapy for publication, he had treated 26 patients, ten of whom had improved dramatically.[101]

The problem with camphor was its unreliability in producing fits. Also, the patients hated the feeling of anxiety preceding the fit, the vomiting that camphor caused, and the pain in the muscles where it was injected. In 1934 the Budapest professor of pharmacology recommended to Meduna a drug that had been synthesized nine years previously and that was being sold

under the name “Cardiazol” as a cardiac stimulant. Meduna began injecting his patients with Cardiazol and by 1936 had tried it on 110 of them, half of whom went into remission (mainly those who had fallen ill more recently).[102]

Cardiazol was never a big success, mainly because it too was unreliable in producing fits and was feared by the patients. Müller soon discarded it at Münsingen because of patients’ “agonizing fears of dying and crumbling away.... It was no wonder that they often strongly resisted a repeat experience.”[103] Indeed, during a Cardiazol fit they looked to be on the point of expiring. Said Müller: “The sight of the artificially produced attack of epilepsy, especially of the contorted blue faces, was so awful to me that I sought to get away from the room whenever I could. I realize now the inadequacy of my excuse that I could have achieved nothing with my presence and that my colleagues were more robust and not so squeamish. But it was my responsibility....”[104] One psychiatrist at a mental hospital in Leicestershire recalled “the unseemly and tragic farce of an unwilling patient being pursued by a posse of nurses with me, a fully charged syringe in my hand, bringing up the rear.”[105]

Walter Freeman had the colossal misjudgment of trying Meduna’s therapy on a 70-year-old “disturbed” relative. He and his surgeon brother injected 6 cc. of Metrazol into one of her veins. “Within 10 seconds she began twitching, then opened her mouth widely, arched her back, stiffened out in a tonic [muscles rigid] convulsion that lasted about 20 seconds, followed by clonic movements [muscles flexing repeatedly] for another 25 seconds. Then she relaxed with no respiratory movements for many seconds. She became cyanotic.... Finally she gasped and then breathed heavily. Gradually, the color returned to her face—and also to my brother’s.”

Applying electroconvulsive therapy (ECT).

Source: The History of Shock Treatment by Leonard Roy Frank (Ed.), 1978, p. 14.

“Jesus!” he said, wiping his brow.

Although the relative remembered Freeman in her will, she never forgave him.[106]

In 1939, Meduna migrated from Budapest to Chicago, where he became a professor first at Loyola University then at the University of Illinois Medical School, engaging in tenebrous experiments with carbon-dioxide therapy.[107] With the onset of war in Europe, Cardiazol therapy, like most of the other innovations in psychiatry, would become an American story. Institutions as disparate as the giant asylum at Milledgeville and the fashionable Sheppard private clinic experimented with Cardiazol (Metrazol) units in the late 1930s and early 1940s.[108] Then the hated Metrazol was pushed into the dustbin of history with the advent of a much less detested form of convulsive therapy, electroconvulsive therapy, or ECT.

Electroshock

It was now 1938. The air was full of talk about the new "physical therapies," as they had begun to be called, distinct from psychotherapy and psychoanalysis. Barbiturate narcosis, insulin coma, and Metrazol convulsion had burst with full force into the life of the asylum, extending the promise of lasting remissions and even cures. The next step was electroconvulsive therapy, which Ugo Cerletti, professor of psychiatry in Rome, used for the first time in April 1938. Although the history of psychiatry had known many previous applications of electricity, Cerletti's historic innovation was giving electric shock to the brain to achieve a convulsion.[109]

Cerletti was born in 1877 in the small manufacturing town of Conegliano, about 80 miles from Venice, where his father founded the first school of viticulture in Italy. He began medical studies in 1896 in Turin, transferring two years later to Rome to work in the neuropathology laboratory of Giovanni Mingazzini. While still a medical student, Cerletti traveled to Heidelberg to study brain histology with Nissl. Finishing his MD in 1901, Cerletti became an assistant physician of the psychiatric clinic in Rome. Until World War I, he continued to do postgraduate work in Germany and France, studying for example with Kraepelin at the university clinic in Munich. He soon established for himself a substantial reputation as an investigator of the architecture of the brain.[110] Up to this point, little distinguished Cerletti from other typical representatives of the first biological psychiatry in its dying days.

After World War I, Cerletti received a series of academic psychiatric postings and had ample opportunity to experience clinical psychiatry firsthand as a "funereal science."[111] In these years, particularly during his stint in the early 1930s as director of the Neuropsychiatric Clinic in Genoa, he did research on epilepsy. Was a lesion in a certain area of the brain the cause or the result of an epileptic attack? To study the question, Cerletti began inducing fits in dogs experimentally by applying electricity to them.[112] Yet with one electrode placed in the animal's mouth, the other in its anus, the results were intimidating: Half of the animals died as a result of shock stopping the action of the heart. In 1935, Cerletti received the professorship of psychiatry in Rome, becoming simultaneously chief of the Clinic for Nervous and Mental Diseases of the University of Rome.

In October 1936, he called together his three assistants, Ferdinando Accornero, Lucio Bini, and Lamberto Longhi, themselves all young MDs in their mid-20s and at the end of residencies.[113] Impressed by Meduna's recent results with Cardiazol, Cerletti gave each of his assistants an assignment: Accornero was to study insulin coma, Longhi was to investigate Cardiazol, and Bini was to see if the dog experiments might be made applicable to humans. The three young psychiatrists were inseparable comrades, together day and night in the Rome clinic, and functioned more as a team than as individual researchers off in separate laboratories. It was, however, Bini who discovered that electric current could be delivered safely if the electrodes were placed on the dog's temples.[114] For a year, these dog experiments continued, the dog-catcher's wagon stopping weekly at the clinic, Accornero and Bini scrambling around the clock to perform the autopsies on the animals' brains and to get slides of brain sections under the microscope.

In 1937, Max Müller's big conference at Münsingen on new therapies in schizophrenia began to loom, and Accornero and Bini consulted with Cerletti about what they should present there. Accornero would offer the results of his insulin research and Bini would give a paper on the dog experiments. It was decided that Bini would casually mention that the dog experiments might be tried on humans. (He did mention it; the news was received at the conference with complete indifference.[115])

After the two had returned to Rome, Cerletti asked them to visit the city slaughterhouse, where, he had heard, pigs were being obtunded with electric current before being slaughtered. What could the researchers learn here about where to apply the electrodes, and about the margin between the convulsive dose and the lethal dose? Bini and Accornero then did systematic experiments at the slaughterhouse, discovering that placing the electrodes on the temples was practical, and that the margin between the convulsive and lethal doses was very wide.

Now all three young residents were keen to try electroshock on a patient. Still, Cerletti hesitated. Said Accornero later, "If the experiment on a human, for whatever unforeseen reason, had ended with the death of the subject, all the responsibility would have fallen on Cerletti. Our school, already considered to be highly interventionist, would have fallen into enormous discredit and its director would have suffered the consequences." [116] In the meantime, however, with the assistance of a technician, Bini had constructed a primitive apparatus that would permit application of 80 to 100 volts of

electricity for a fraction of a second. Now the first patient arrived.

On April 15, 1938, the Rome commissariat of police referred to the clinic a 39-year-old engineer from Milan who had been arrested while wandering about the railroad station. “He does not appear to be in full possession of his mental faculties,” said the commissar’s note, “and I am sending him to your hospital to be kept there under observation.” In clinic, the patient seemed lucid and knew where he was, but spoke in a kind of jargon and complained about being “telepathically influenced.” He had hallucinations. His grooming was slovenly. “From a psychiatric viewpoint,” said young resident Accornero, “the syndrome was clear, the disorder advanced, the presumed prognosis unfavorable.”[117]

On the second floor *(primo piano)* of the clinic, there was an isolated equipment room. There Bini had set up his apparatus. On that morning of April 18, there were present in the room, in addition to the patient, Cerletti, Bini, Accornero, and two other assistant physicians, one assigned to keep watch on the corridor and make sure no one would intrude. The patient, his head shaved, seemed quite indifferent to what was going on. A nurse placed the electrodes on his temples while an orderly put a rubber tube between his teeth to prevent him from biting his tongue. Everything was ready. Bini looked at Cerletti, who nodded. There was a crack of electricity. The patient’s muscles jolted once. Accornero put his stethoscope on the patient’s chest. Heart rate up, everything else alright. Accornero was so excited he found himself unable to speak.

The patient said he had no memory of what had just happened. “We gave him 80 volts for a tenth of a second,” said Bini. “He had an absence attack.”

“Let’s step it up to 90,” said Cerletti.

Another electrical crack. Another spasm. The patient lay motionless for a minute, then began to sing.

“We’ll try it one last time at a higher voltage,” said Cerletti, “poi *basta* [and then enough].”

At this point, the patient said, in a perfectly calm and reasonable voice, as though answering an exam question, “Look out! The first is pestiferous, the second mortiferous.” The residents looked at each other puzzled.

“Come on, let’s go,” said Cerletti.

Bini turned the apparatus up to the maximum. After this jolt, the patient went into a classic tonic-clonic epileptic fit, muscles contracting and relaxing rhythmically. His breathing stopped. He began turning blue, heart

accelerating, corneal reflexes absent.

Bini began counting the seconds since the halting of respiration. At the forty-eighth second, the patient emitted a profound sigh. The doctors likewise. They had established that electrical current could safely produce convulsions in a human subject.

The patient sat up “calm and smiling, as though to inquire what we wanted of him,” as Cerletti later remembered.

They asked, “What happened to you?”

He answered, “I don’t know. Maybe I was asleep.”

Thus ended the first application of electricity to produce a convulsion in a psychiatric patient.[118] Cerletti at once baptized it “electroshock.”[119]

After eleven applications of ECT, the patient in fact did get well and was discharged from the clinic a month later “in good condition and well-oriented; ideation and memory perfect.” He now realized that his previous notions of persecution and hallucinations were the result of illness. He returned to his job in Milan, and a year later was said still to be “perfectly well; but his wife reported that three months after his return home he resumed his jealous attitude toward her, and that sometimes during the night he would speak as though in answer to voices.”[120]

Thus ECT was not a cure for schizophrenia. But it represented a great alleviation of the disabling symptoms of psychotic illness, and permitted individuals to function more or less normally. It spread with great rapidity within the world of psychiatry. Lothar Kalinowsky functioned as a kind of Johnny Appleseed. Born in Berlin with one Jewish parent, Kalinowsky earned an MD at the University of Berlin in 1922, then fled Germany in 1933 for Italy, where he acquired a second MD in Rome. He was present at the Cerletti clinic in the early days of ECT therapy. Leaving Rome in 1939 for Paris, he helped to introduce ECT at the Ste-Anne Hospital (obliging Bini to send instructions for building a machine). Arriving in England in July 1939, Kalinowsky helped Sanderson McGregor establish ECT at the Netherne Hospital at Coulsdon.[121] A German colleague at St. Bartholomew’s Hospital asked Kalinowsky to attend rounds and had an apparatus constructed.[122] Thus Kalinowsky sowed ECT across the land.

In March 1940, Kalinowsky, an alien physician, reluctantly left England for the United States and by September 1940 got an ECT service going at the New York State Psychiatric Institute, which was part of Columbia University. They built their own machine.[123]

Everywhere the electroshock machine was introduced, it occasioned great enthusiasm among asylum psychiatrists. One no longer had to chase patients down to inject them with Metrazol. “ECT produced instant unconsciousness, no dread, no physical upset after the convulsion, no vomiting,” said one Bethlem physician, who recalled the arrival of the ECT apparatus at that hospital. “It was the size of a small cinema organ, its top a rather bewildering array of dials and switches. Although the machine was fearsome to us nurses and doctors to begin with, the fears of patients about convulsion treatment became unusual and outright refusal rare.”[124] As another veteran psychiatrist declared, “Without ECT I would not have lasted out in psychiatry, as I would not have been able to tolerate the sadness and hopelessness of most mental illnesses before the introduction of convulsive therapy.”[125] By 1944 in England, ECT had supplanted Cardiazol in producing fits.[126]

The priority for bringing ECT to the United States belongs technically not to Kalinowsky—who did insert it in academic psychiatry—but to Renato Almansi, who had worked with Cerletti and who in 1939 imported a machine into the United States. Almansi collaborated with David Impastato, also Italian by origin but who had an American MD degree and who was on staff at the Columbus Hospital in New York. They first did experiments on dogs, then by February 1940 had begun applying ECT to patients at their office and at the hospital outpatient department.[127] Elsewhere in the country in these months, other attempts were simultaneously underway and no one person save Kalinowsky emerges as the godparent of American ECT.[128]

Yet ECT did not have a truly triumphal onmarch in the United States.[129] For one thing, the psychoanalysts were generally opposed to it, even though many of them conceded its usefulness in the treatment of major depression (where up to a quarter of all patients suicided). Washington psychoanalyst Harry Stack Sullivan had no more use for ECT than for any of the other physical therapies, based as they were (in his view) on the philosophy “that it is better to be a contented imbecile than a schizophrenic.” Sullivan thought it better to leave schizophrenic patients untreated in the hope that these “extraordinarily gifted and, therefore, socially significant people” would experience a spontaneous recovery.[130] In 1947, the Group for the Advancement of Psychiatry bristled about the “promiscuous and indiscriminate use” of ECT while not condemning it outright.[131] (The organization later backpedaled somewhat to maintain they opposed only its

“inappropriate” use.)[132] Throughout the American psychoanalytic literature ran a begrudging suspiciousness of electroshock. This was combined with an insistence that ECT’s obvious utility must rest on some kind of psychodynamic basis rather than on brain biology.[133] For if the neurons of the brain itself were making people ill, the theoretical structure of psychoanalysis flew out the window.

Thus ECT posed psychoanalysis with a dilemma. The most effective means of treating major mental illness was theoretically unacceptable. As a resident, one would train either to be a psychotherapist, with a large and lucrative community practice, or a poorly paid mental-hospital psychiatrist who was able to do ECT.[134] When Arnold Rogow interviewed a cross-section of American psychiatrists and psychoanalysts in 1966, he found that a third of the psychiatrists used ECT but only one of the analysts.[135] The hesitant acceptance of ECT in the United States in the late 1940s and 1950s was thus partially due to ideological opposition from psychiatrists who found the physical therapies outside their worldview.

But it was also owing to real dangers in the use of ECT itself. As the patients thrashed upon the table, they were at risk of breaking limbs and fracturing vertebrae. At the Horton Hospital in England, the nurses would literally drape themselves on the patient to limit mobility, one holding the feet together, another pressing down on the pelvis, two more nurses on each side holding the patient’s shoulders with one hand and the patient’s hand with the other, and another nurse holding the patient’s head and compressing the jaw.[136]

These dangers began to be mastered as a result of a chance encounter around 1939 between Walter Freeman and Omaha psychiatrist Abram Bennett. Bennett had been giving children with spastic paralyses small amounts of curare, attempting to relax their limbs by blocking the junction between muscle and nerve. He had also been conducting Metrazol shock therapy on depressed patients. Freeman suggested to Bennett that curare might solve the problem of fractured spinal vertebrae in shock therapy. Freeman knew a businessman with good curare connections in Ecuador, and so Bennett got a large supply of curare, which was standardized in the pharmacology laboratory of the University of Nebraska to make its effects consistent and predictable.[137] This represented curare’s introduction into medicine.

In 1940, Bennett established that curare in fact briefly paralyzed the

muscles of head and neck, preventing thrashing.[138] His discovery was soon applied to ECT, allowing the extension of electroshock to large numbers of patients with depression. Yet curare was a highly risky drug and could entail cardiac complications. A safer blocker of the nerve-muscle junction, succinylcholine, had been introduced into medicine in 1949 for short procedures such as inserting a tracheal tube. Starting in 1952, it was adopted in ECT units as a less risky means of preventing spinal fractures.[139] It would be used together with the ultrashort-acting barbiturate methohexital sodium (trademark "Brevital"), which served as a general anesthetic.

By 1959, ECT had become "the treatment of choice," as Kalinowsky put it, for manic-depressive illness and major depression.[140] It was more effective than any of the other physical therapies, acted swiftly, and was not unpopular with the patients. The year 1959 was a kind of golden year for psychiatry, when neither Kalinowsky nor anyone else knew that the antipsychiatry movement was about to bring ECT to an end.

In retrospect, the shock therapies represented a milestone in breaking psychiatry free from the tutelage of neurology. Until the 1930s, most of the psychiatric spectrum had been dominated by neurologists with their virtual monopoly on psychotherapy in private practice and their command of the few procedures, such as spa therapy for nerves, that seemed to do any good. Until then, psychiatry had been a poor Cinderella, eking out a paltry existence in the asylum. Malaria therapy, deep sleep, and the shock treatments represented the first independent therapies over which psychiatrists themselves disposed. Sakel, Meduna, and Cerletti succeeded in "shaking neurology from the saddle," as Louis Casamajor, a senior New York psychiatrist, put it in 1943. Added Casamajor tongue in cheek, "One may question whether shock treatments do any good to the patients but there can be no doubt that they have done an enormous amount of good to psychiatry."[141]

Walter Freeman, a neurologist at George Washington University Hospital in Washington D.C., who popularized lobotomy in the United States. Shown here in 1949 inserting a lobotomy instrument under the patient's eyelid in order to destroy tissue in the brain's frontal lobe (courtesy of UP1/Corbis-Bettmann).

The Lobotomy Adventure

The idea of operating on the brain to cure madness does not seem intrinsically unreasonable. Physicians have always intuited that a physical intervention in the brain, perhaps cutting some tract causing compulsive behavior or removing a center producing some malignant protein, might put an end to a pattern of psychosis. In the Middle Ages, doctors fantasized about cutting for the mythical "stone of madness." In our own time, there is evidence that the course of Parkinson's disease can be slowed by the transplantation of fetal dopamine-producing neurons.[142] It also seems to be the case that ablating the cingulate gyrus, in a cingulotomy, can relieve the symptoms of severe compulsive neurosis.[143] These are examples of successful psychosurgery.

The great wave of physical therapies of the 1930s included one unsuccessful example of psychosurgery: lobotomy, or destroying part of one of the brain's lobes. In the context of the apparent success of such physical interventions as barbiturate narcosis, insulin coma, Metrazol convulsion, and ECT, it began to seem reasonable that operating on the cerebral cortex itself might offer promise. And so psychosurgical notions last seen in the 1880s and 1890s began to resurface.

The modern history of psychosurgery begins with a Swiss asylum psychiatrist named Gottlieb Burckhardt, who earned his MD at Basel in 1860. After a lectureship at Basel, followed by a staff job at the Waldau public asylum, in 1882 Burckhardt became director of the private Préfargier Clinic in Marin near Neuchâtel.[144] Here he seems to have experimented widely with various new techniques such as hypnotism.[145] Untrained though he was in surgery, he also experimented rather clumsily with operations on various lobes of his patients' brains. Beginning in December, 1888, he performed psychosurgery on six patients with evident schizophrenia. The operations were not a big success. One of the patients died in convulsions, a second improved, a third and fourth experienced no change, and the last two became "quieter."[146] Burckhardt's report on his new procedure to the Berlin Medical Congress of 1890 produced disquiet. There was no discussion following the paper, and the participants agreed that complete silence was the best way to avoid attracting attention to the work.[147] Burckhardt's article the following year did occasion a brief flutter.[148] But the idea of intervening directly in the parenchyma of the brain was soon abandoned.

Yet a climate of meddlesomeness persisted. In the 1890s, there were efforts in England to relieve the symptoms of neurosyphilis by boring a burr hole (trepanation) in the cranium and incising the meninges to drain away pus or cerebrospinal fluid.[149] In unpublished work, the Breslau surgeon Johann Mikulicz attempted in the 1890s to remove the focus of epilepsy by scarifying a portion of the sensory-motor cortex.[150] Valentin Magnan, master of the Ste-Anne asylum in Paris, was enthusiastic about trepanation to forestall mental retardation in cases where the cranium was said to be too small. (He alluded to previous efforts along these lines, which he found "unencouraging.")[151] Thus there is no doubt that at the turn of the century a generalized willingness to meddle surgically with the brain and its casing was ticking away just beneath the surface of medicine. It required only a new

finding, or pseudo-finding, to reactivate this meddlesomeness.

At this point, the Lisbon neurologist Egas Moniz walks on stage. Moniz had already earned himself a place in medical history by describing in 1927 cerebral angiography, a technique for visualizing the blood vessels of the brain radiographically. He had twice been nominated for a Nobel prize for this achievement and twice turned down. In this frame of mind of thirsting for greatness denied, Moniz attended the Second International Congress of Neurology held in London in 1935. Here he audited a full-day symposium on the frontal lobes of the brain, then an object of great interest among many scholars. Moniz heard Carlyle Jacobsen and John Fulton from Yale describe the emotional changes in a chimpanzee following the ablation of much of its frontal lobes: Prone to temper tantrums and wilfullness before the operation, the animal seemed to become almost cheery after it.

"Dr. Moniz arose and asked, if frontal lobe removal prevents the development of experimental neuroses in animals and eliminates frustrational behavior, why would it not be feasible to relieve anxiety states in man by surgical means?"[152]

The inspiration for lobotomy, or leucotomy as Moniz called it, had been planted. Between November 1935 and February 1936, Moniz prevailed upon neurosurgeon Almeida Lima to resect part of the prefrontal lobes of 20 patients transferred from the Bombarda asylum to the neurology service of the Santa Marta Hospital in Lisbon. Seven had been "cured," seven ameliorated, and in six there was no change.

Moniz provided little detail to support these claims.[153] In fact, this kind of grandiose communication characterized the first notice of almost all the physical therapies, their authors eager to ascertain their priority in history and unaffected as yet by the rigorous statistical tests and follow-up studies that would later be demanded. (Sakel's first report in the Vienna medical press did not even mention the number of patients receiving the insulin-coma therapy.) Yet with lobotomy, one was dealing with a savage mutilation of the human brain.

It was therefore most unfortunate that Washington neurologist Walter Freeman, who also had attended the London conference, fell upon Moniz's procedure with uncritical enthusiasm. For it was Freeman who, in collaboration with neurosurgeon James Watts, propagandized on behalf of lobotomy in the United States. (The term "lobotomy" is theirs.) In 1936, they did their first operation at the hospital of George Washington University in

Washington, and in 1946 introduced the transorbital lobotomy, approaching the brain from the roof of the orbit, or eye socket.[154]

Freeman and Watts split in 1947 over the issue of making lobotomy an "office procedure," as Freeman wished it. Thereafter Freeman carried on alone. In the early 1950s, he popularized transorbital lobotomy in a kind of one-man medicine show traveling across the continent. In the summer of 1951, for example, he visited mental hospitals in 17 states as well as doing demonstrations in Canada, Puerto Rico, and Curaçao. In the words of Freeman's biographer Elliot Valenstein, "On one five-week summer trip that year, he drove 11,000 miles with a station wagon loaded, in addition to camping equipment, with an electroconvulsive shock box, a dictaphone, and a file cabinet filled with patient records, photographs, and correspondence; his surgical instruments were in his pocket."[155] The surgical instruments often included an ice pick.

Although lobotomy did tend to tranquilize the raving patients who were management problems, it generally deprived them of their judgment and social skills, making them tin-eared to social cues and inappropriately disinhibited. The authors of the standard textbook of physical therapies of the day wrote the following lines apologetically: "It is probable that every individual after the operation is happier than before, but this may be bought at too great a cost, not only to himself but to society....

"The patient's temper becomes more hasty; he will be more easily irritable and will more easily give vent to his irritation, sometimes without adequate sense of its social consequences. He may become demanding in his desires and imperative in their expression."[156] What the authors did not say was that much of the patient's selfhood as well was given up with the physical loss of part of the brain's frontal lobes, producing, in the words of two of Freeman's opponents, "a defrontalized dement."[157]

There were two main forms of lobotomy. Moniz preferred prefrontal leucotomy, cutting the white matter in the oval center of the brain's two front lobes with a whisk-like leukotome through burr-holes cut in the top of the cranium. When stirred back and forth, the leukotome would destroy the nerve fibers. The version that Freeman propagated after 1946 was transorbital lobotomy, going via the orbital cavity.

Here Dr. Hatcher of the Milledgeville State Hospital describes the transorbital procedure on one February morning in 1952 to psychologist Peter Cranford:[158]

DR. HATCHER:	"Peter, I'm doing transorbital lobotomies this morning. Come watch me."
CRANFORD:	"If I *saw* one, you'd have to do the next one on me."
DR. HATCHER:	"Nothing to it. I take a sort of medical icepick, hold it like this, bop it through the bones just above the eyeball, push it up into the brain, swiggle it around, cut the brain fibers like this, and that's it. The patient doesn't feel a thing."
CRANFORD:	"And neither do you. I *was* going to breakfast but I've changed my mind."
HATCHER *(laughing)*:	"You can change your mind, but not like I can change it."[158]

In the 1950s, the world of American asylum psychiatry was Milledgeville writ large. Although only 684 lobotomies were conducted in the United States between 1940 and 1944, the propagandizing of Freeman and Watts helped the numbers soar in the late 1940s. In 1949, the peak year, 5,074 operations were carried out. "By 1951," writes historian Gerald Grob, "no fewer than 18,608 individuals had undergone psychosurgery since its introduction in 1936."[159] It was employed at more than half of the public mental hospitals in the country, and such prestigious institutions as the Boston Psychopathic Hospital counted as fortresses of lobotomy.

The procedure faded away in the early 1950s almost as abruptly as it had risen up, a blip in the history of psychiatry, though an illuminating one as a study in medical hubris. In Britain, the rate of lobotomies began to decline even before the introduction in the mid-1950s of the new antipsychotic drugs.[160] But in both Britain and the United States, it was unquestionably the advent of these drugs in the spring of 1954 that killed off lobotomy,[161] just as other physical therapies of the 1930s also died in the face of the new psychopharmacology.

In retrospect, frontal lobotomy was indefensible for ethical reasons. Yet it did have some success with agitated patients whom other physical therapies were unable to reach, or on whom they had not been tried. "After having personally performed perhaps 12 or 15 prefrontal lobotomies (all carefully selected and screened)," said one psychiatrist in 1987, "there were some

gratifying results in people who had been institutionalized and uncontrollable for as long as 6 to 10 years. There were some dramatic responses, and some long-standing satisfactory responses—all better than the restraint of those same patients in back wards of mental institutions." [162] Follow-up studies found that about a third of all psychosurgical patients had been discharged from hospital and were living at home.[163] Yet many of these patients would sooner or later have recovered spontaneously. And the irreversible damage to their brain and spirit must be weighed against the extra months or years with which they would have encumbered the institutional system. "Not all so-called mental disorders were so severe that it was worth exchanging them for an organic brain syndrome," concluded one student.[164] True, lobotomy reached the most difficult of the difficult in the back wards. Yet unlike any of the other physical therapies, it caused deep uneasiness within the profession of psychiatry, and would be the first of these therapies to be abandoned as the new antipsychotic drugs came in.

Social and Community Psychiatry

These physical therapies represented a series of alternatives to the dilemma of custodialism versus psychoanalysis. Yet there was one further alternative. It concerned the milieu of therapy rather than a physical approach to the body. Social and community psychiatry insisted it was not the patients' genes nor early childhood that made them ill but the surrounding community. Psychiatric illness could thus best be addressed with therapies aimed at placing patients in a healing community environment. This tends to be an English story.

Each country has made its own distinctive contribution to the history of psychiatry. Germany offered the first biological psychiatry, France the therapeutic asylum. The United States contributed psychoanalysis in its fullest bloom and, latterly, much of the second biological psychiatry. The distinctive English contribution to the world narrative was the notion that at the bottom of mental illness lay poisoned human relationships. If psychosis and neurosis arose as a result of faulty human connectivity, illness could be treated by restoring healthy relationships to a person's life, principally in the form of group therapy.

The whole story of social and community psychiatry as a unified approach to mental illness is like a Lego, or Tinkertoy set, in that its various components, or pieces, had lain about the psychiatric landscape for many years. Only in Britain around the time of World War II did they come together in a single edifice.

One such piece was the open asylum. In an open asylum, voluntary patients could admit and discharge themselves at will, an essential concept if psychotic illness were to be destigmatized and neurotic illness made accessible to the psychiatrist. Germany had a long tradition of such institutions in the private sector, beginning with Otto Müller's Clinic for Nervous Patients, founded in 1861 in Helmstedt (and moved in 1865 to scenic Blankenburg in the Harz Mountains). As for the private asylum, in 1866 Adolf Albrecht Erlenmeyer established in his insane asylum in Bendorf-on-Rhine an open division that patients could enter and leave relatively at will.[165]

Another piece of the Lego set was discharging patients to some kind of family-based community care, not necessarily that of their own families. Putting patients in family care had been done for many years in the famous Belgian "colony for the insane" at Gheel, yet its example seems to have influenced few others. Scottish asylums were well-known for their policies of family care.[166] In the mid-nineteenth century, the Germans made extensive use of placing mental patients as boarders in the homes of nearby farm families eager to pick up a bit of spending money. From 1867 on, Caspar Max Brosius, director of a private asylum in Bendorf, started placing his wealthy patients under supervised care in private homes, thus avoiding administrative formalities and the ensuing stigma.[167] Ferdinand Wahrendorff later tried the same approach at a private asylum in Ilten near Hanover on a larger and more systematic scale.[168] By the turn of the century, "family care" *(Familienpflege)* had become common in Central Europe in both public and private sectors.

Outpatient clinics based in the asylum also increased the porousness of the mental hospital to the surrounding community. By the mid-1920s, most of the big Central European psychiatric hospitals had such outreach clinics. Charcot had founded an outpatient clinic at the Salpêtrière in Paris. By 1920, Massachusetts had 33 outpatient clinics, New York 25, and Pennsylvania 9.[169] Psychiatric outpatient clinics in general hospitals took this integration another step further. The first in Germany was created at Rostock in 1825.[170]

Thus along an entire continuum of care, moves had been made before the 1930s to erase the boundary between the closed asylum and the community. The destruction of German psychiatry after 1933 erased the memory of most of these initiatives. In the two decades between 1930 and 1950, it was Britain that vaulted from international laggard to international leader.

Why this transformation took place is unclear, a mixture perhaps of burgeoning social democracy from below and the activism from above of socially committed elites. (One recalls how the Bloomsbury group flung themselves into psychoanalysis.) The transformation began in 1930 with the passage of a lunacy act of a decidedly different tenor. The Mental Treatment Act of 1930 began to open the asylum to the outside community, reversing completely the course set by the Lunacy Act of 1890, which sealed the "insane" in madhouses like dangerous beasts.[171] The new legislation made possible open-door policies in asylums, discarding locks and keys, and enabling patients on "outside parole" to stroll at will into town for a pint. (There were three other kinds of parole, which, rather than facilitating mass flight, had a tranquilizing influence on the inhabitants of the mental hospitals.[172]) Among the first to implement open-door policies was the "kind and benevolent Welshman" T. P. ("Percy") Rees, who on becoming superintendent in 1935 of Warlingham Park Hospital ordered that the gates be unlocked. All the ward doors in the hospital were unlocked too, including, to the horror of the staff, those of the "suicidal" patients.[173]

Yet the crucial event in the genesis of social and community psychiatry in Britain was not the 1930 Act but World War II. Following the Nazi invasion of Austria in 1938, Joshua Bierer, a 37-year-old Jewish physician and psychotherapist, fled from Vienna to London. Bierer had studied under Alfred Adler and had already acquired much psychotherapeutic experience in private practice and at the Steinhof mental hospital in Vienna. Once installed in 1938 at Runwell Hospital near London, an ultramodern facility organized on the "villa" system, he proceeded to set up a psychotherapy program. This was quite unusual in British mental hospitals then, few of which had full-time psychotherapists. Bierer asked his patients about their dreams and their earliest recollections from childhood.[174]

What was extraordinary however in the context of prewar psychiatry was group psychotherapy. In 1939, Bierer began to organize psychotherapy groups among Runwell inpatients, also for outpatients at two public general hospitals in London. In 1942, he extended group psychotherapy to the

historic London teaching hospitals, Guy's and St. Bartholomew's.[175] Bierer thought that psychoanalysis prolonged the patients' problems by making them dependent on an analyst. Group psychotherapy by contrast let the patients become "independent, active and 'self-deciding,'" while at the same time helping them to achieve insight and work toward a cure.[176] This marks the first eruption of group psychotherapy in Britain and possibly on the continent of Europe.

How did these groups first assemble? "On December 8, 1939, 35 patients, neurotics and psychotics, met in 'Sunnyside House' [of Runwell Hospital] and formed a social club. The chair at the meeting was taken by a patient." The therapeutic group, or therapeutic community as it would soon be called, was in other words run by the patients themselves and not imposed on them top-down by the hospital staff. A staff member would be present at Social Club meetings, but otherwise the Club was "completely autonomous," the patients electing their own officers, publishing a magazine, and meeting three times weekly. A typical discussion question might be, "Why do we laugh about the downfall of others?"[177] "The discipline was left entirely in the hands of Club members," said Bierer. Thus Bierer had not merely facilitated group psychotherapy but established the principle that it be patient directed. He referred to the technique as " 'community' treatment."[178]

Now the scene shifts from Bierer to a northern suburb of London, where at the outbreak of war the Ministry of Health set up the Mill Hill Emergency Hospital, a psychiatric center located on the site of a former public school, for the treatment of military and civilian shell-shock cases. It was staffed by members of the Maudsley Hospital and included an Effort Syndrome Unit, so named because much psychosomatic illness among troops presented itself in the form of shortness of breath and the like following exercise (so-called soldier's heart). Heading this 100-bed unit were a cardiologist and a psychiatry resident, Maxwell Jones, a young Scottish physician (born in South Africa) who had graduated several years previously from the University of Edinburgh and was at the Maudsley at the time the war broke out. It was under Jones at Mill Hill that the notion of "therapeutic community" truly germinated. And it germinated almost by accident. Many of the war nurses were not old-fashioned authoritarian ward sisters, but mature professional women who had chosen nursing as their war work. These women were accustomed to free communication, not to giving orders to patients and receiving them from doctors. A nurse might "frequently keep a

ward log book," said Jones, "where the fourteen patients in her ward recorded some of the problems affecting the group, and described how these problems were met through discussion." So under these middle-class volunteer nurses some kind of group dynamic among the patients began to take form.

At the same time at Mill Hill, the doctors would bring the patients together in a group to explain, in a nice way, that what they really had was a form of hysteria. No particular therapeutic benefit was expected from this group congregation at first: It was merely an efficient way to communicate information without alienating the patients as a lecture might have. "It soon became evident," continued Jones, "that the discussion group was more than an educational meeting; it was affecting the whole social structure of the ward." Patients began raising problems arising from ward life. The atmosphere of the group would flux in unpredictable ways. The nurses who attended the meetings said the same process was occurring on their wards. Doctors and nurses at Mill Hill began to reflect on what these changes might mean. They began experimenting by raising social problems in the groups, or having the nurses act out little skits in front of the patients involving "the lives of a fictitious family comprising the parents (the 'normal' father and hysterical mother) and three daughters whose personalities tended to be schizoid, psychopathic, and hysterical respectively. This dramatic approach proved to be tremendously interesting to the patients and provoked a high degree of participation in the subsequent discussion." By January 1944, the patients themselves had begun to participate in the psychodramas, though as yet they did not use that word because they had not yet heard of Jacob Moreno, who first used psychodrama in an asylum.[179] Thus by 1944, Jones was toying with the possibilities of group interactions in and of themselves as a source of therapeutic benefit. He did not use the term group psychotherapy.

Early in 1945, the therapeutic community was field-tested on a larger scale. The Mill Hill group was asked to take over a unit at the Southern Hospital at Dartford in Kent for the treatment of returning prisoners of war with combat fatigue. Six psychiatrists, 50 nurses, and support staff from Mill Hill now applied their therapeutic model to this 300-bed service, helping the men find day-work in Dartford, to which they would commute from the hospital in three large Green Line buses. As Jones explained, "The men were housed in six 'cottages,' each with 50 beds. Each unit had a daily community meeting, along the lines we had developed at Mill Hill." In this kind of supportive atmosphere, the patients would discuss such matters as their fears

about resuming sex with their wives, the children born while they were away, and their adequacy as husbands.[180]

Dartford led to a "considerable growth of interest in therapeutic communities," as Jones put it. Funded by various ministries, in April 1947 Jones opened what would become the most famous therapeutic community in the world, a 100-bed unit to study the problem of "the chronic unemployed neurotic" at the newly named Belmont Hospital, the dilapidated old wing of a London County Council asylum. (During the war many of the staff of the Maudsley Hospital who had not gone to Mill Hill set up a neurosis center at Belmont more oriented to the physical therapies than the psychological ones.[181]) Jones's new service was called the Industrial Neurosis Unit; it put into action all the concepts that he and his team had been developing. The patients' week was filled with group activity:

> Monday: Unit conference, "when the patients air their grievances or make constructive suggestions." Tuesday: Enlightening films.
>
> Wednesday-Thursday: Unit staff member leads discussion groups. Friday: Psychodrama. Also patients do occupational therapy from 10 to 12 and 2 to 4. Early Friday evening a pass outside, then at 7 P.M. an organized social program got up by the patients. "This is readily censured at the group discussions should it fail to cater for all needs."[182]

The Industrial Neurosis Unit became in 1954 the Social Rehabilitation Unit, later the Henderson Hospital (named after the Scottish psychiatrist David Henderson who had influenced many of these reformers), a premier center of social and community psychiatry.

The therapeutic community was an alternative to psychoanalysis on the one hand and custodialism on the other. It was not an alternative to the physical therapies, for even though the Industrial Neurosis Unit was not intended to accept psychotic patients, in fact it often did so. Jones and his staff would regularly do insulin coma treatment. They used sodium amytal, a barbiturate, for interviewing patients, and occasionally performed lobotomy on the premises.[183]

However, let's not get ahead of our story. Therapeutic communities became widely heralded during the war. Thomas Main, a former deputy superintendent of an asylum in Northumbria, set one up at the Northfield Military Hospital near Birmingham, emphasizing a kind of instantaneous

emotional contact among staff themselves and between staff and patients. This became known as “the Second Northfield Experiment” and Main claimed credit for actually introducing the term “therapeutic community.”[184] Northfield, said Main, was “a therapeutic setting with a spontaneous and emotionally structured (rather than medically dictated) organization in which all staff and patients engage.”[185] “Sincerity,” he explained, was the “basis for management.”[186] Although Main’s account of his activities was permeated with psychoanalytic jargon, it is interesting that at Northfield they gave barbiturate narcosis and ECT.[187]

The reason these progressive ideas disseminated so rapidly within the usually stodgy military medical bureaucracy was that they had some very powerful sponsors, far higher up the ladder than Jones and his associates. Main was one of what were known as the “Tavi brigadiers,” a group of psychiatrists from the Tavistock Clinic in central London, who had taken charge of army psychiatry. The Tavistock Clinic had been founded in 1920 by Hugh Crichton-Miller and John Rawlings (“J. R.”) Rees as an outpatient service having a quasi-psychoanalytic orientation for patients with nervous illnesses. In 1933, Rees became its medical director. At the outbreak of war in 1939, Rees became the director of army psychiatry. He brought with him into the military Research and Training Center a number of bright young psychoanalytically oriented psychiatrists, one of whom was Main. Another was John Bowlby, later noted for his work on attachment theory. As early as 1942, Rees was using the term “social psychiatry,” a concept that had existed since Esquirol but a term that only at this point was gaining currency. From early in the war on, a number of seminars at the Tavistock dealt with social themes.[188]

Yet the therapeutic community needed a home. The doctrine emphasized the empowerment of patients and the normalizing of their lives, good community relations making well what bad human relations had caused.[189] The asylum therefore was not a perfect home for the group psychotherapy that was at the core of the therapeutic community. Where else then to situate it? Here was born the notion of the psychiatric day hospital, a kind of outpatient department attached to a mental hospital, a general hospital, or even free-standing, where patients could come for group sessions, counseling, occupational therapy and the other services that constituted a comprehensive approach to treatment. It was actually Ewen Cameron who established the first day hospital in 1946 at the Allan Memorial Institute in

Montreal (introducing the term day hospital into psychiatry as well).[190] But Joshua Bierer was unaware of Cameron's efforts at the time, and in 1948 Bierer established the first day hospital in England.

Its home was the Social Psychiatry Centre that Bierer had founded in 1946 in two war-damaged houses in the London suburb of Hampstead. He and coworkers soon began discussing the outlines of a possible day hospital that would combine a patients' Social Club with psychodrama, group psychotherapy, ECT and insulin cure, and the other innovations that had established themselves within progressive British psychiatry. Over the next 12 months, support for the concept welled among hospital administrators, psychiatrists in private practice, and family doctors. By October 1948, Bierer had lined up funding for eight psychiatrists on a part-time basis, two full-time psychiatrists, a part-time psychologist, a psychiatric social worker, an occupational therapist, a social therapist, and other staff.[191] This represented the apogee of British psychiatry's golden age, a tremendous marshaling of resources that was conceivable only in a social setting in which "social and community psychiatry" had become a watchword.

The day hospital caught on at once as a humane and inexpensive alternative to institutional care. Additional day hospitals were founded at Bristol in 1951, at the Maudsley in 1953, and so forth. By 1959, more than 38 such hospitals existed in Great Britain.[192] The movement incorporated Bierer's philosophy that, "Treatment must include the whole social environment of the patient and all his social relationships. He must be treated not only as a person but as part of a community."[193]

In retrospect, the day hospital movement is significant as the first effort to move the treatment of major psychiatric disorders from the asylum to the community. However naive this policy might seem with the benefit of hindsight, in the world of postwar Britain, Jones's therapeutic communities and Bierer's "social and community psychiatry" loomed as exciting prospects, true alternatives to the pessimism of the neurobiological approach on the one hand and to the esoteric rites of psychoanalysis on the other.

There was much less excitement in the United States. Although the Americans used such terms as social psychiatry, they usually meant some form of psychiatric epidemiology or study of human relations in the mental hospital rather than group psychotherapy, the core of social and community psychiatry in Britain.

Around the time of World War I, American psychiatry was hit by a strong

activist impulse, trying to break the asylum stalemate and to see the patient and his illness in social terms. Among the first to use the term "social psychiatry" was Elmer Southard in 1917, director of the Boston Psychopathic Hospital, who understood a mixture of psychiatric social services and social psychology.[194] In the early 1920s, a string of speakers at annual meetings of the American Psychiatric Association, including two of the Association's presidents, gave impassioned addresses on the asylum's need to reach out to the community or to forestall mental illness through prevention in the community.[195] This kind of rhetoric was evidence of psychiatry's willingness to break out of the custodial mode (see pp. 67-68) and return to the mainstream of medicine. It fit with the American doctrine of "social medicine" in the 1920s, seeing medicine in part as social science and directing attention to "the whole man."[196] None of this had anything to do with group psychotherapy or community care.

Group psychotherapy made its initial eruption in the United States in 1934, as Paul Schilder at Bellevue in New York started bringing groups of two to seven patients in the outpatient department together for collective sessions once or twice a week. "In group psychotherapy," said Schilder, "a number of patients are seen simultaneously by the physician, and each patient is aware of the problems of the others." The patients, many of whom were quite disturbed, experienced relief at hearing they were not alone with their difficulties. "In one discussion, for example, a patient remembered an attempted sexual assault against his sister. It was astonishing how many members of the group recalled similar experiences in their own lives, so that a correct appreciation of such an event became possible." [197] The inventive Schilder, however, would soon be dead in an automobile accident. Unlike Britain, where the military establishment lay behind group psychotherapy, the movement did not take off in the United States. In the United States, psychoanalysts seized control of army psychiatry during World War II, and Freud's doctrines rather than patients' social clubs took pride of place.

The term social psychiatry thrived in the United States during the 1950s and 1960s, in the sense of large surveys of the mental health of the population such as the Midtown Manhattan Study, commissioned in 1950 by Thomas Rennie of Cornell Medical College.[198] Under "therapeutic communities," Americans understood mainly mental hospitals with open-door policies, a serious mistranslation of an Anglicism. And the American definition of social psychiatry was broad enough to encompass every influence imaginable on

the patient's life: childhood, friends, social class, poverty, and the like. All of these elements helped give American psychiatry its stamp of social activism but bore little in common with what was understood in Britain under the term: group psychotherapy, day hospitals, and patient autonomy.

Later, therapeutic communities did flourish in the United States. "No progressive hospital worth its salt would be without the label of therapeutic community," said one observer in the early 1970s.[199] Yet these experiments were frequently deformed by the spirit of psychoanalysis with its emphasis on a one-to-one relationship between doctor and patient (to achieve "transference"). When the Americans did attempt to form a therapeutic community, they often produced a caricature of the British original. Witness the nightmare community that Ken Kesey described in his novel, *One Flew Over the Cuckoo's Nest* (1962) (see p. 275). There was one day hospital in the United States: at the Menninger Clinic.[200]

American "community psychiatry" ended in tragedy, the massive disorgement of disabled asylum patients to the rough care of the streets (see pp. 280-281). Little resembling the day care offered in Britain or on the Continent was available in the United States. The 1963 mental health act of John F. Kennedy's administration might be seen as an exception to this generalization. Yet the mental health centers the act made possible soon were given over to psychotherapy for middle-class neurotics rather than to the community care of patients with psychotic illness.[201] The subject of community psychiatry in the United States was, and remains, a kind of grotesque joke.

Yet for the grand narrative of psychiatry's history, the failure of communitarianism to gain headway in the United States did nudge events along. The absence of a sensible rival doctrine to psychoanalysis permitted biological psychiatry to overthrow the competition more easily than was the case in Europe, where community psychiatry offered tough resistance for decades to the biological approach.[202] But then, biological and social perspectives are not necessarily incompatible: Psychiatric illness is often triggered and molded by social stresses. Recognition of this gave social psychiatry a much more tenacious hold than the analysts possessed.

For half a century, the discipline of psychiatry stood trapped between the choices of custodial care and individual psychoanalysis. Forced to come up with alternatives, psychiatry cobbled together a band of options ranging from bromide sleep to ECT to social clubs. Among these alternative choices, there

were no conflicting paradigms, no theoretical battles. A doctor might order psychodrama one day and ECT the next. So desperate were workaday psychiatrists to dodge the awful choice between custodialism (for the poor) and psychoanalysis (for the rich), that they were willing to try anything that held promise. Thus for half a century, the discipline of psychiatry averted choosing between the neurobiological paradigm on the one hand and the psychogenic paradigm on the other. After the 1960s, this kind of pragmatic eclecticism no longer became possible. The neurobiological paradigm came roaring back from the grave in which Kraepelin had interred it—with medications that truly worked and evidence that psychiatric illness represented a biological phenomenon far deeper than troubled human relations or a schizophrenogenic mother. The alternatives of the first half of the twentieth century were almost all wiped from the slate by the advent of the second biological psychiatry.